

CHAPTER

8.2

ELECTROSTATICS

1. Let $Q_1 = 4 \mu\text{C}$ be located at $P_1(3, 11, 8)$ while $Q_2 = -5 \mu\text{C}$ is at $P_2(6, 15, 8)$. The force F_2 on Q_2 will be
(A) $-(4.32\mathbf{u}_x + 5.76\mathbf{u}_y)$ N (B) $4.32\mathbf{u}_x + 5.76\mathbf{u}_y$ N
(C) $-(4.32\mathbf{u}_x + 5.76\mathbf{u}_y)$ mN (D) $4.32\mathbf{u}_x + 5.76\mathbf{u}_y$ mN
2. Four 5 nC positive charge are located in the $z=0$ plane at the corners of a square 8 mm on a side. A fifth 5 nC positive charge is located at a point 8 mm distant from each of the other charge. The magnitude of the total force on this fifth charge is
(A) 2×10^{-4} N (B) 4×10^{-4} N
(C) 0.014 N (D) 0.01 N
3. Four 40 nC are located at $A(1, 0, 0)$, $B(-1, 0, 0)$, $C(0, 1, 0)$ and $D(0, -1, 0)$ in free space. The total force on the charge at A is
(A) $24.6\mathbf{u}_x \mu\text{N}$ (B) $-24.6\mathbf{u}_x \mu\text{N}$
(C) $-13.6\mathbf{u}_x \mu\text{N}$ (D) $13.76\mathbf{u}_x \mu\text{N}$
4. Let a point charge 41 nC be located at $P_1(4, -2, 7)$ and a charge 45 nC be at $P_2(-3, 4, -2)$. The electric field $\mathbf{E}$ at $P_3(1, 2, 3)$ will be
(A) $0.13\mathbf{u}_x + 0.33\mathbf{u}_y + 0.12\mathbf{u}_z$
(B) $-0.13\mathbf{u}_x - 0.33\mathbf{u}_y - 0.12\mathbf{u}_z$
(C) $1.15\mathbf{u}_x + 2.93\mathbf{u}_y + 1.09\mathbf{u}_z$
(D) $-1.15\mathbf{u}_x - 2.93\mathbf{u}_y - 1.09\mathbf{u}_z$
5. Let a point charge 25 nC be located at $P_1(4, -2, 7)$ and a charge 60 nC be at $P_2(-3, 4, -2)$. The point, at which on the y axis, is $E_x = 0$, is
(A) -7.46 (B) -22.11
(C) -6.89 (D) (B) and (C)
6. A $2 \mu\text{C}$ point charge is located at A (4, 3, 5) in free space. The electric field at P(8, 12, 2) is
(A) $131.1\mathbf{u}_\rho + 159.7\mathbf{u}_\phi - 49.4\mathbf{u}_z$
(B) $159.7\mathbf{u}_\rho + 27.4\mathbf{u}_\phi - 49.4\mathbf{u}_z$
(C) $131.1\mathbf{u}_\rho + 27.4\mathbf{u}_\phi - 49.4\mathbf{u}_z$
(D) $159.7\mathbf{u}_\rho + 137.1\mathbf{u}_\phi - 49.4\mathbf{u}_z$
7. A point charge of -10 nC is located at $P_1(0, 0, 0.5)$ and a charge of $2 \mu\text{C}$ at the origin. The $\mathbf{E}$ at P(0, 2, 1) is
(A) $68.83\mathbf{u}_r + 14.85 \mathbf{u}_\phi$ (B) $68.83\mathbf{u}_r + 64.01\mathbf{u}_\phi$
(C) $68.83\mathbf{u}_r - 14.85 \mathbf{u}_\phi$ (D) $68.83\mathbf{u}_r - 64.01\mathbf{u}_\phi$
8. Charges of 20 nC and -20 nC are located at (3, 0, 0) and $(-3, 0, 0)$ and $(-3, 0, 0)$, respectively. The magnitude of $\mathbf{E}$ at y axis is
(A) $\frac{1080}{(9 + y^2)^{3/2}}$ (B) $\frac{1080}{(9 + y^2)^3}$
(C) $\frac{108}{(9 + y^2)^{3/2}}$ (D) $\frac{108}{(9 + y^2)^3}$
9. A charge Q_0 located at the origin in free space produces a field for which $E_2 = 1$ kV/m at point P(-2, 2, -1). The charge Q_0 is
(A) $2 \mu\text{C}$ (B) $-3 \mu\text{C}$
(C) $3 \mu\text{C}$ (D) $-2 \mu\text{C}$
10. The volume charge density $\rho_v = \rho_o e^{-|x| - |y| - |z|}$ exist over all free space. The total charge present is
(A) $2\rho_o$ (B) $4\rho_o$
(C) $8\rho_o$ (D) $3\rho_o$

11. A uniform volume charge density of $0.2 \mu\text{C}/\text{m}^2$ is present throughout the spherical shell extending from $r = 3 \text{ cm}$ to $r = 5 \text{ cm}$. If $\rho = 0$ elsewhere, the total charge present throughout the shell will be

- (A) 41.05 pC (B) 257.92 pC
(C) 82.1 pC (D) 129.0 pC

12. If $\rho_v = \frac{1}{z^2 + 10} 5e^{-0.1\rho}(\pi - |\phi|) \mu\text{C}/\text{m}^3$ in the region $0 \leq \rho \leq 10$, $-\pi < \phi < \pi$ and all z , and $\rho_v = 0$ elsewhere, the total charge present is

- (A) 1.29 mC (B) 2.58 mC
(C) 0.645 mC (D) 0

13. The region in which $4 < r < 5$, $0 < \theta < 25^\circ$, and $0.9\pi < \phi < 1.1\pi$ contains the volume charge density of $\rho_v = 10(r - 4)(r - 5) \sin \theta \sin \frac{\phi}{2}$. Outside the region, $\rho_v = 0$. The charge within the region is

- (A) 0.57 C (B) 0.68 C
(C) 0.46 C (D) 0.23 C

14. A uniform line charge of $5 \text{ nC}/\text{m}$ is located along the line defined by $y = -2$, $z = 5$. The electric field $\mathbf{E}$ at $P(1, 2, 3)$ is

- (A) $-9\mathbf{u}_y + 45\mathbf{u}_z$ (B) $9\mathbf{u}_y - 45\mathbf{u}_z$
(C) $-18\mathbf{u}_y + 9\mathbf{u}_z$ (D) $18\mathbf{u}_y - 9\mathbf{u}_z$

15. A uniform line charge of $6.25 \text{ nC}/\text{m}$ is located along the line defined by $y = -2$, $z = 5$. The $\mathbf{E}$ at that point in the $z = 0$ plane where the direction of $\mathbf{E}$ is given by $(\frac{1}{3}\mathbf{u}_y - \frac{2}{3}\mathbf{u}_z)$, is

- (A) $45\mathbf{u}_y + 9\mathbf{u}_z$ (B) $45\mathbf{u}_y - 9\mathbf{u}_z$
(C) $9\mathbf{u}_y - 18\mathbf{u}_z$ (D) $18\mathbf{u}_y - 36\mathbf{u}_z$

16. Uniform line charge of $20 \text{ nC}/\text{m}$ and $-20 \text{ nC}/\text{m}$ are located in the $x = 0$ plane at $y = 3$ and $y = -3 \text{ m}$ respectively. The $\mathbf{E}$ at $P(6, 0, 6)$ will be

- (A) $-24\mathbf{u}_y \text{ V}/\text{m}$ (B) $48\mathbf{u}_y \text{ V}/\text{m}$
(C) $-48\mathbf{u}_y \text{ V}/\text{m}$ (D) $24\mathbf{u}_y \text{ V}/\text{m}$

17. Uniform line charges of $100 \text{ nC}/\text{m}$ lie along the entire extent of the three coordinate axes. The $\mathbf{E}$ at $P(-3, 2, -1)$ is

- (A) $-1.92\mathbf{u}_x + 2\mathbf{u}_y - 1.08\mathbf{u}_z \text{ kV}/\text{m}$
(B) $-0.96\mathbf{u}_x + \mathbf{u}_y - 0.54\mathbf{u}_z \text{ kV}/\text{m}$
(C) $0.96\mathbf{u}_x - \mathbf{u}_y + 0.54\mathbf{u}_z \text{ kV}/\text{m}$
(D) $1.92\mathbf{u}_x - 2\mathbf{u}_y + 1.08\mathbf{u}_z \text{ kV}/\text{m}$

18. Two identical uniform charges with $\rho_l = 80 \text{ nC}/\text{m}$ are located in free space at $x = 0$, $y = \pm 3 \text{ m}$. The force per unit length acting on the line at positive y arising from the charge at negative y is

- (A) $9.375\mathbf{u}_y \mu\text{N}$ (B) $37.5\mathbf{u}_y \mu\text{N}$
(C) $19.17\mathbf{u}_y \mu\text{N}$ (D) $75\mathbf{u}_y \mu\text{N}$

19. A uniform surface charge density of $10 \text{ nC}/\text{m}^2$ is present in the region $x = 0$, $-2 < y < 2$ and all z if $\epsilon = \epsilon_0$, the electric field at $P(3, 0, 0)$ has

- (A) x component only
(B) y component only
(C) x and y component
(D) x , y and z component

20. The surface charge density is $\rho_s = 5 \text{ nC}/\text{m}^2$, in the region $\rho < 0.2$, $z = 0$, and is zero elsewhere. The electric field $\mathbf{E}$ at $A(\rho = 0, z = 0.5)$ is

- (A) $5.4 \text{ V}/\text{m}$ (B) $10.1 \text{ V}/\text{m}$
(C) $10.5 \text{ V}/\text{m}$ (D) $20.2 \text{ V}/\text{m}$

21. Three infinite charge sheet are positioned as follows: $10 \text{ nC}/\text{m}^2$ at $x = -3$, $-40 \text{ nC}/\text{m}^2$ at $y = 4$ and $50 \text{ nC}/\text{m}^2$ at $z = 2$. The $\mathbf{E}$ at $(4, 3, -2)$ is

- (A) $0.56\mathbf{u}_x + 2.23\mathbf{u}_y - 2.8\mathbf{u}_z \text{ kV}/\text{m}$
(B) $0.56\mathbf{u}_x - 2.23\mathbf{u}_y + 2.8\mathbf{u}_z \text{ kV}/\text{m}$
(C) $0.56\mathbf{u}_x + 2.23\mathbf{u}_y + 2.8\mathbf{u}_z \text{ kV}/\text{m}$
(D) $-0.56\mathbf{u}_x - 2.23\mathbf{u}_y + 2.8\mathbf{u}_z \text{ kV}/\text{m}$

22. Let $\mathbf{E} = 5x^3\mathbf{u}_x - 15x^2y\mathbf{u}_y$. The equation of the stream line that passes through $P(4, 2, 1)$ is

- (A) $y = \frac{128}{x^3}$ (B) $x = \frac{128}{y^3}$
(C) $y = \frac{64}{x^2}$ (D) $x = \frac{64}{y^2}$

23. A point charge 10 nC is located at origin. Four uniform line charge are located in the $x = 0$ plane as follows : $40 \text{ nC}/\text{m}$ at $y = 1$ and -5 m , $-60 \text{ nC}/\text{m}$ at $y = -2$ and -4 m . The $\mathbf{D}$ at $P(0, -3, 4)$ is

- (A) $-19.1\mathbf{u}_y + 25.5\mathbf{u}_z \text{ pC}/\text{m}^2$
(B) $19.1\mathbf{u}_y - 25.5\mathbf{u}_z \text{ pC}/\text{m}^2$
(C) $-16.4\mathbf{u}_y + 21.9\mathbf{u}_z \text{ pC}/\text{m}^2$
(D) $16.4\mathbf{u}_y - 21.9\mathbf{u}_z \text{ pC}/\text{m}^2$

24. A point charge 20 nC is located at origin. Four uniform line charge are located as follows 40 nC/m at $y = \pm 1$ and 50 nC/m at $y = \pm 2$. The electric flux that leaves the surface of a sphere, 4 m in radius, centered at origin is

- (A) 1.33 nC (B) 1.89 μ C
(C) 1.33 μ C (D) 1.89 μ C

25. The cylindrical surface $\rho = 8$ C contains the surface charge density, $\rho_s = 5e^{-20|z|}$ nC/m². The flux that leaves the surface $\rho = 8$ cm, $1 \text{ cm} < z < 5 \text{ cm}$ $30^\circ < \phi < 90^\circ$ is

- (A) 270.7 nC (B) 9.45 nC
(C) 270.7 pC (D) 9.45 pC

26. Let $\mathbf{D} = 4xy \mathbf{u}_x + 2(x^2 + z^2)\mathbf{u}_y + 4yz\mathbf{u}_z$ C/m². The total charge enclosed in the rectangular parallelepiped $0 < x < 2$, $0 < y < 3$, $0 < z < 5$ m is

- (A) 360 C (B) 180 C
(C) 100 C (D) 560 C

27. Volume charge density is located in free space as $\rho_v = 2e^{-1000r}$ nC/m³ for $0 < r < 1$ mm and $\rho_v = 0$ elsewhere. The value of D_r on the surface $r = 1$ mm is

- (A) 1.28 pC/m² (B) 0.28 pC/m²
(C) 0.78 pC/m² (D) 0.32 pC/m²

28. Spherical surfaces at $r = 2$ and 4 carry uniform charge densities of 20 nC/m² and -4 nC/m². The D_r at $2 < r < 4$ is

- (A) $-\frac{16}{r^2}$ nC/m² (B) $\frac{16}{r^2}$ nC/m²
(C) $\frac{80}{r^2}$ nC/m² (D) $-\frac{80}{r^2}$ nC/m²

29. Given the electric flux density, $\mathbf{D} = 2xy\mathbf{u}_x + x^2\mathbf{u}_y + 6z^3\mathbf{u}_z$ C/m². The total charge enclosed in the volume $0 < x, y, z < a$ is

- (A) $6a^5 + \frac{5}{3}a^4$ (B) $a^5 + 6a^4$
(C) $6a^5 + a^4$ (D) $\frac{5}{3}a^5 + 6a^4$

30. Let $\mathbf{D} = 5x^4y^4z^4\mathbf{u}_y$. The flux enclosed by volume $x = 3$ and 3.1, $y = 1$ and 1.1, and $z = 2$ and 2.1 is

- (A) 49.6 (B) 24.8
(C) 35.4 (D) 36.4

31. A spherical surface of radius of 3 mm is centered at P(4, 1, 5) in free space. If $\mathbf{D} = x\mathbf{u}_x$ C/m² the net electric flux leaving the spherical surface is

- (A) 113.1 μ C (B) 339.3 nC
(C) 113.1 nC (D) 452.4 nC

32. The electric flux density is

$$\mathbf{D} = \frac{1}{z^2} [10xyz\mathbf{u}_x + 5x^2z\mathbf{u}_y + (2z^3 - 5x^2y)\mathbf{u}_z]$$

The volume charge density ρ_v at $(-2, 3, 5)$ is

- (A) 6.43 (B) 8.96
(C) 10.4 (D) 7.86

33. If $\mathbf{D} = 2r\mathbf{u}_r$ C/m², the total electric flux leaving the surface of the cube, $0 < x, y, z < 0.4$ is

- (A) 0.32 (B) 0.34
(C) 0.38 (D) 0.36

34. If $\mathbf{E} = 4\mathbf{u}_x - 3\mathbf{u}_y + 5\mathbf{u}_z$ in the neighborhood of point P(6, 2, -3). The incremental work done in moving 5 C charge a distance of 2 m in the direction $\mathbf{u}_x + \mathbf{u}_y + \mathbf{u}_z$ is

- (A) -60 J (B) 34.64 J
(C) -34.64 J (D) 60 JJ

35. If $\mathbf{E} = 100\mathbf{u}_\rho$ V/m, the incremental amount of work done in moving a 60 μ C charge a distance of 2 mm from P(1, 2, 3) toward Q(2, 1, 4) is

- (A) -5.4 μ J (B) 3.1 μ J
(C) -3.1 μ J (D) 0

36. A 10 C charge is moved from the origin to P(3, 1, -1) in the field $\mathbf{E} = 2x\mathbf{u}_x - 3y^2\mathbf{u}_y + 4\mathbf{u}_z$ V/m along the straight line path $x = -3y$, $y = x + 2z$. The amount of energy required is

- (A) -40 J (B) 20 J
(C) -20 J (D) 40 J

37. A uniform surface charge density of 30 nC/m² is present on the spherical surface $r = 6$ mm in free space. The V_{AB} between A($r = 2$ cm, $\theta = 35^\circ$, $\phi = 55^\circ$) and B($r = 3$ cm, $\theta = 40^\circ$, $\phi = 90^\circ$) is

- (A) 2.03 V (B) 10.17 V
(C) 4.07 mV (D) -10.17 V

38. A point charge is located at the origin in free space. The work done in carrying a charge 10 C from point A ($r = 4$, $\theta = \pi/6$, $\phi = \pi/4$) to B ($r = 4$, $\theta = \pi/3$, $\pi/6$) is

- (A) $0.45 \mu\text{J}$ (B) $0.32 \mu\text{J}$
(C) $-0.45 \mu\text{J}$ (D) 0

39. Let a uniform surface charge density of 5 nC/m^2 be present at the $z = 0$ plane, a uniform line charge density of 8 nC/m be located at $x = 0$, $z = 4$ and a point charge of $2 \mu\text{C}$ be present at P(2, 0, 0). If $V = 0$ at A(0, 0, 5), the V at B(1, 2, 3) is

- (A) 10.46 kV (B) 1.98 kV
(C) 0.96 kV (D) 3.78 kV

40. A non uniform linear charge density, $\rho_L = 6/(z^2 + 1) \text{ nC/m}$ lies along the z axis. The potential at P($\rho = 1$, 0, 0) in free space is ($V_\infty = 0$)

- (A) 0 V (B) 216 V
(C) 144 V (D) 108 V

41. The annular surface, $1 \text{ cm} < \rho < 3 \text{ cm}$ carries the nonuniform surface charge density $\rho_s = 5\rho \text{ nC/m}^2$. The V at P(0, 0, 2 cm) is

- (A) 81 mV (B) 90 mV
(C) 63 mV (D) 76 mV

42. If $V = 2xy^2z^3 + 3\ln(x^2 + 2y^2 + 3z^2)$ in free space the magnitude of electric field E at P (3, 2, -1) is

- (A) 72.6 V/m (B) 79.6 V/m
(C) 75 V/m (D) 70.4 V/m

43. It is known that the potential is given by $V = 70r^{0.6} \text{ V}$. The volume charge density at $r = 0.6 \text{ m}$ is

- (A) 1.79 nC/m^3 (B) -1.79 nC/m^3
(C) 1.22 nC/m^3 (D) -1.22 nC/m^3

44. The potential field $V = 80r^2 \cos \theta \text{ V}$. The volume charge density at point P(2.5, $\theta = 30^\circ$, $\phi = 60^\circ$) in free space is

- (A) -2.45 nC/m^3 (B) 1.42 nC/m^3
(C) -1.42 nC/m^3 (D) 2.45 nC/m^3

45. Within the cylinder $\rho = 2$, $0 < z < 1$ the potential is given by $V = 100 + 50\rho + 150\rho \sin \phi \text{ V}$. The charge lies within the cylinder is

- (A) -4.94 nC (B) -4.86 nC
(C) -5.56 nC (D) -3.68 nC

46. A dipole having

$$\frac{Qd}{4\pi\epsilon_0} = 100 \text{ V} \cdot \text{m}^2$$

is located at the origin in free space and aligned so that its moment is in the $\mathbf{u}_z$ direction. The E at point ($r = 1$, 45° , $\phi = 0$) is

- (A) 158.11 V/m (B) 194.21 V/m
(C) 146.21 V/m (D) 167.37 V/m

47. A dipole located at the origin in free space has a moment $p = 2 \times 10^{-9} \mathbf{u}_z \text{ C} \cdot \text{m}$. The points at which $|E|_0 = 1 \text{ mV/m}$ on line $y = z$, $x = 0$ are

- (A) $y = z = \pm 23.35$ (B) $y = z = \pm 16.5$, $x = 0$
(C) $y = z = 16.5$ (D) $y = 0$, $z = 23.35$, $x = 0$

48. A dipole having a moment $\mathbf{p} = 3\mathbf{u}_x - 5\mathbf{u}_y + 10\mathbf{u}_z \text{ nC} \cdot \text{m}$ is located at P(1, 2, -4) in free space. The V at Q (2, 3, 4) is

- (A) 1.31 V (B) 1.26 V
(C) 2.62 V (D) 2.52 V

49. A potential field in free space is expressed as $V = 40/xyz$. The total energy stored within the cube $1 < x, y, z < 2$ is

- (A) 1548 pJ (B) 0
(C) 774 pJ (D) 387 pJ

50. Four 1.2 nC point charge are located in free space at the corners of a square 4 cm on a side. The total potential energy stored is

- (A) $1.75 \mu\text{J}$ (B) $2 \mu\text{J}$
(C) $3.5 \mu\text{J}$ (D) 0

51. Given the current density

$$\mathbf{J} = 10^5 [\sin(2x) e^{-2y} \mathbf{u}_x + \cos(2x) e^{-2y} \mathbf{u}_y] \text{ kA/m}^2$$

The total current crossing the plane $y = 1$ in the $\mathbf{u}_y$ direction in the region $0 < x < 1$, $0 < z < 2$ is

- (A) 0 (B) 12.3 mA
(C) 24.6 mA (D) 6.15 mA

- 64.** In a certain region where the relative permittivity is 2.4, $\mathbf{D} = 2\mathbf{u}_x - 4\mathbf{u}_y + 5\mathbf{u}_z$ nC/m². The polarization is
- (A) $2.8\mathbf{u}_x - 5.6\mathbf{u}_y + 7\mathbf{u}_z$ nC/m²
 (B) $3.4\mathbf{u}_x - 6.9\mathbf{u}_y + 8.6\mathbf{u}_z$ nC/m²
 (C) $1.2\mathbf{u}_x - 2.3\mathbf{u}_y + 2.9\mathbf{u}_z$ nC/m²
 (D) $3.89\mathbf{u}_x - 6.43\mathbf{u}_y + 8.96\mathbf{u}_z$ nC/m²

- 65.** Medium 1 has the electrical permittivity $\epsilon_1 = 1.5\epsilon_0$ and occupies the region to the left of $x=0$ plane. Medium 2 has the electrical permittivity $\epsilon_2 = 2.5\epsilon_0$ and occupies the region to the right of $x=0$ plane. If $\mathbf{E}_1$ in medium 1 is $\mathbf{E}_1 = (2\mathbf{u}_x - 3\mathbf{u}_y + 1\mathbf{u}_z)$ V/m then $\mathbf{E}_2$ in medium 2 is
- (A) $(2.0\mathbf{u}_x - 1.8\mathbf{u}_x + 0.6\mathbf{u}_z)$ V/m
 (B) $(1.67\mathbf{u}_x - 3\mathbf{u}_y + \mathbf{u}_z)$ V/m
 (C) $(1.2\mathbf{u}_x - 3\mathbf{u}_y + \mathbf{u}_z)$ V/m
 (D) $(1.2\mathbf{u}_x - 1.8\mathbf{u}_y + 0.6\mathbf{u}_z)$ V/m

- 66.** Two perfect dielectrics have relative permittivities $\epsilon_{r1} = 2$ and $\epsilon_{r2} = 8$. The planar interface between them is the surface $x - y + 2z = 5$. The origin lies in region 1. If $\mathbf{E}_1 = 100\mathbf{u}_x + 200\mathbf{u}_y - 50\mathbf{u}_z$ V/m then $\mathbf{E}_2$ is
- (A) $400\mathbf{u}_x + 800\mathbf{u}_y - 200\mathbf{u}_z$ V/m
 (B) $400\mathbf{u}_x + 200\mathbf{u}_y - 50\mathbf{u}_z$ V/m
 (C) $25\mathbf{u}_x + 200\mathbf{u}_y - 50\mathbf{u}_z$ V/m
 (D) $125\mathbf{u}_x + 175\mathbf{u}_z$ V/m

- 67.** The two spherical surfaces $r = 4$ cm and $r = 9$ cm are separated by two perfect dielectric shells, $\epsilon_{r1} = 2$ for $4 < r < 6$ and $\epsilon_{r2} = 5$ for $6 < r < 9$. If $\mathbf{E}_1 = \frac{1000}{r^2} \mathbf{u}_r$ then $\mathbf{E}_2$ is
- (A) $\frac{5000}{r^2} \mathbf{u}_r$ V/m
 (B) $\frac{400}{r^2} \mathbf{u}_r$ V/m
 (C) $\frac{2500}{r^2} \mathbf{u}_r$ V/m
 (D) $\frac{2000}{r^2} \mathbf{u}_r$ V/m

- 68.** The surface $x=0$ separate two perfect dielectric. For $x > 0$, let $\epsilon_{r1} = 3$, while $\epsilon_{r2} = 5$ where $x < 0$. If $\mathbf{E}_1 = 80\mathbf{u}_x - 60\mathbf{u}_y - 40\mathbf{u}_z$ V/m then $\mathbf{E}_2$ is
- (A) $(133.3\mathbf{u}_x - 100\mathbf{u}_z - 66.7\mathbf{u}_z)$ V/m
 (B) $(133.3\mathbf{u}_x - 60\mathbf{u}_z - 40\mathbf{u}_z)$ V/m
 (C) $(48\mathbf{u}_x - 36\mathbf{u}_y - 24\mathbf{u}_z)$ V/m
 (D) $(48\mathbf{u}_x - 60\mathbf{u}_y - 40\mathbf{u}_z)$ V/m

- 69.** A potential field exists in a region where $\epsilon = f(x)$. If $\rho_v = 0$, the $\nabla^2 V$ is
- (A) $-\frac{1}{f(x)} \frac{df}{dx} \frac{\partial V}{\partial x}$
 (B) $f(x) \frac{df}{dx} \frac{\partial V}{\partial x}$
 (C) $\frac{1}{f(x)} \frac{df}{dx} \frac{\partial V}{\partial x}$
 (D) $-f(x) \frac{df}{dx} \frac{\partial V}{\partial x}$

- 70.** If $V(x, y) = 4e^{2x} + f(x) - 3y^2$ in a region of free space where $\rho_v = 0$. It is known that both E_x and V are zero at the origin. The $V(x, y)$ is
- (A) $3(x^2 - y^2)$
 (B) $3(y^2 - x^2)$
 (C) $4x^2 - 3y^2$
 (D) $4y^2 - 3x^2$

SOLUTIONS

$$\begin{aligned}
 1. \text{ (C) } \mathbf{F}_2 &= \frac{Q_1 Q_2}{4\pi\epsilon_0 |\mathbf{R}_{12}|^3} \mathbf{R}_{12} \\
 &= \frac{(4 \times 10^{-6})(-5 \times 10^{-6})}{4\pi\epsilon_0} \times \frac{(3\mathbf{u}_x + 4\mathbf{u}_y)}{5^3} \\
 &= (4.32\mathbf{u}_x + 5.76\mathbf{u}_y) \text{ mN}
 \end{aligned}$$

2. (D) Arranging the charge in the xy plane at location $(4, 4)$, $(-4, 4)$, $(4, -4)$, $(-4, -4)$, the fifth charge will be on the z -axis at location $z = 4\sqrt{2}$. By symmetry, the force on the fifth charge will be z directed, and will be four times the z component

$$\begin{aligned}
 F &= \frac{4}{\sqrt{2}} \times \frac{q^2}{4\pi\epsilon_0 d^2} \\
 &= \frac{4}{\sqrt{2}} \times \frac{(5 \times 10^{-9})^2}{4\pi \times (8.85 \times 10^{-12})(8 \times 10^{-3})^2} = 10^{-2} \text{ N}
 \end{aligned}$$

3. (D) The force will be

$$F = \frac{(40 \times 10^{-9})^2}{4\pi\epsilon_0} \left[\frac{\mathbf{R}_{CA}}{|\mathbf{R}_{CA}|^3} + \frac{\mathbf{R}_{DA}}{|\mathbf{R}_{DA}|^3} + \frac{\mathbf{R}_{BA}}{|\mathbf{R}_{BA}|^3} \right]$$

where $\mathbf{R}_{CA} = \mathbf{u}_x - \mathbf{u}_y$, $\mathbf{R}_{DA} = \mathbf{u}_x + \mathbf{u}_y$, $\mathbf{R}_{BA} = 2\mathbf{u}_x$
 $|\mathbf{R}_{CA}| = |\mathbf{R}_{DA}| = \sqrt{2}$, $|\mathbf{R}_{BA}| = 2$

$$\begin{aligned}
 \mathbf{F} &= \frac{(40 \times 10^{-9})^2}{4\pi \times (8.85 \times 10^{-9})} \left[\frac{\mathbf{u}_x - \mathbf{u}_y}{2\sqrt{2}} + \frac{\mathbf{u}_x + \mathbf{u}_y}{2\sqrt{2}} + \frac{2\mathbf{u}_x}{8} \right] \\
 &= 13.76\mathbf{u}_x \text{ } \mu\text{N}
 \end{aligned}$$

$$4. \text{ (C) } \mathbf{E} = \frac{10^{-9}}{4\pi\epsilon_0} \left[\frac{41\mathbf{R}_{13}}{|\mathbf{R}_{13}|^3} + \frac{45\mathbf{R}_{23}}{|\mathbf{R}_{23}|^3} \right]$$

$$\begin{aligned}
 \mathbf{R}_{13} &= -3\mathbf{u}_x + 4\mathbf{u}_y - 4\mathbf{u}_z, \quad \mathbf{R}_{23} = 4\mathbf{u}_x - 2\mathbf{u}_y + 5\mathbf{u}_z \\
 \mathbf{E} &= 9 \times 10^9 \times 10^{-9} \times \left[\frac{41(-3\mathbf{u}_x + 4\mathbf{u}_y - 4\mathbf{u}_z)}{(41)^{3/2}} + \frac{45(4\mathbf{u}_x - 2\mathbf{u}_y + 5\mathbf{u}_z)}{(45)^{3/2}} \right] \\
 &= 1.152\mathbf{u}_x + 2.93\mathbf{u}_y + 1.089\mathbf{u}_z
 \end{aligned}$$

5. (D) The point is $P_3(0, y, 0)$

$$\mathbf{R}_{13} = -4\mathbf{u}_x + (y+2)\mathbf{u}_y - 7\mathbf{u}_z,$$

$$\mathbf{R}_{12} = 3\mathbf{u}_x + (y-4)\mathbf{u}_y + 2\mathbf{u}_z$$

$$E_x = \frac{10^{-9}}{4\pi\epsilon_0} \left[\frac{25 \times (-4)}{[65 + (y+2)^2]^{3/2}} + \frac{60 \times 3}{[13 + (y-4)^2]^{3/2}} \right]$$

To obtain $E_x = 0$, $0.48y^2 + 13.92y + 73.12 = 0$
 which yields the two value $y = -6.89, -22.11$

$$6. \text{ (B) } \mathbf{E}_p = \frac{2 \times 10^{-6}}{4\pi\epsilon_0} \frac{\mathbf{R}_{AP}}{|\mathbf{R}_{AP}|^3}$$

$$= \frac{2 \times 10^{-6}}{4\pi\epsilon_0} \left[\frac{4\mathbf{u}_x + 9\mathbf{u}_y - 3\mathbf{u}_z}{(106)^{3/2}} \right]$$

$$= 65.9\mathbf{u}_x + 148.3\mathbf{u}_y - 49.4\mathbf{u}_z$$

Then at point P, $\rho = \sqrt{8^2 + 12^2} = 14.4$,

$$\phi = \tan^{-1} \frac{12}{8} = 56.3^\circ, \text{ and } z = 2$$

$$\begin{aligned}
 E_\rho &= \mathbf{E}_p \cdot \mathbf{u}_\rho = 65.9(\mathbf{u}_x \cdot \mathbf{u}_\rho) + 148.3(\mathbf{u}_y \cdot \mathbf{u}_\rho) \\
 &= 65.9 \cos 56.3^\circ + 148.3 \sin 56.3^\circ = 159.7
 \end{aligned}$$

$$\begin{aligned}
 E_\phi &= \mathbf{E}_p \cdot \mathbf{u}_\phi = 65.9(\mathbf{u}_x \cdot \mathbf{u}_\phi) + 148.3(\mathbf{u}_y \cdot \mathbf{u}_\phi) \\
 &= -65.9 \sin 56.3^\circ + 148.3 \cos 56.3^\circ = 27.4,
 \end{aligned}$$

$$E_z = -49.4$$

$$7. \text{ (C) } \mathbf{E}_p = \frac{2 \times 10^{-8}}{4\pi\epsilon_0} \left[-\frac{\mathbf{R}_1}{|\mathbf{R}_1|^3} + \frac{2\mathbf{R}_2}{|\mathbf{R}_2|^3} \right]$$

$$\mathbf{R}_1 = (0, 2, 1) - (0, 0, 0.5) = (0, 2, 0.5)$$

$$\mathbf{R}_2 = (0, 2, 1) - (0, 0, 0) = (0, 2, 1)$$

$$\mathbf{E}_p = 9 \times 10^9 \times 10^{-8} \left[\frac{-(2\mathbf{u}_y + 0.5\mathbf{u}_z)}{(4.25)^{3/2}} + \frac{2(2\mathbf{u}_y + \mathbf{u}_z)}{(\sqrt{5})^{3/2}} \right]$$

$$\mathbf{E}_p = 54.9\mathbf{u}_y + 44.1\mathbf{u}_z$$

$$\text{At } P, r = \sqrt{5}, \theta = \cos^{-1} \frac{1}{\sqrt{5}} = 63.4^\circ \text{ and } \phi = 90^\circ$$

$$\begin{aligned}
 \text{So } E_r &= \mathbf{E}_p \cdot \mathbf{u}_r = 54.9[\mathbf{u}_y \cdot \mathbf{u}_r] + 44.1[\mathbf{u}_z \cdot \mathbf{u}_r] \\
 &= 54.9 \sin \theta \sin \phi + 44.1 \cos \theta = 68.83
 \end{aligned}$$

$$\begin{aligned}
 E_\theta &= \mathbf{E}_p \cdot \mathbf{u}_\theta = 54.9[\mathbf{u}_y \cdot \mathbf{u}_\theta] + 44.1[\mathbf{u}_z \cdot \mathbf{u}_\theta] \\
 &= 54.9 \cos \theta \sin \phi + 44.1(-\sin \theta) = -14.85
 \end{aligned}$$

$$E_\phi = \mathbf{E}_p \cdot \mathbf{u}_\phi = 54.9(\mathbf{u}_y \cdot \mathbf{u}_\phi) + 44.1(\mathbf{u}_z \cdot \mathbf{u}_\phi) = 54.9 \cos \phi = 0$$

8. (A) Let a point on y axis be $P(0, y, 0)$

$$\mathbf{E}_p = \frac{20 \times 10^{-8}}{4\pi\epsilon_0} \left[-\frac{\mathbf{R}_1}{|\mathbf{R}_1|^3} - \frac{\mathbf{R}_2}{|\mathbf{R}_2|^3} \right]$$

$$\mathbf{R}_1 = (0, y, 0) - (3, 0, 0) = (-3, y, 0)$$

$$\mathbf{R}_2 = (0, y, 0) - (-3, 0, 0) = (3, y, 0),$$

$$|\mathbf{R}_1| = |\mathbf{R}_2| = \sqrt{9 + y^2}$$

$$\mathbf{E}_p = 20 \times 10^{-9} \times 9 \times 10^9 \left[\frac{-3\mathbf{u}_x + y\mathbf{u}_y}{(\sqrt{9 + y^2})^3} + \frac{3\mathbf{u}_x + y\mathbf{u}_y}{(\sqrt{9 + y^2})^3} \right]$$

$$= \frac{-1080\mathbf{u}_x}{(9 + y^2)^{3/2}}, \quad |\mathbf{E}| = \frac{1080}{(9 + y^2)^{3/2}}$$

9. (B) The field at P will be

$$\mathbf{E}_p = \frac{Q_0}{4\pi\epsilon_0} \left[\frac{-2\mathbf{u}_x + 2\mathbf{u}_y - \mathbf{u}_z}{9^{3/2}} \right], \quad E_z = 1 \text{ kV/m}$$

$$Q_0 = -4\pi\epsilon_0 \times 9^{3/2} \times 10^3 = -3 \mu\text{C}$$

10. (C) This will be 8 times the integral of ρ_v over the first octant

$$Q = 8 \int_0^\infty \int_0^\infty \int_0^\infty \rho_o e^{-x-y-z} dx dy dz = 8\rho_o$$

$$\begin{aligned} 11. (C) \quad Q &= \int_0^{2\pi} \int_0^\pi \int_{0.03}^{0.05} 0.2r^2 \sin \theta \, dr d\theta d\phi \\ &= \left[4\pi(0.2) \frac{r^3}{3} \right]_{0.03}^{0.05} = 82.1 \text{ pC} \end{aligned}$$

$$\begin{aligned} 12. (A) \quad Q &= \int_{-\infty}^\infty \int_{-\pi}^\pi \int_0^{10} \frac{5e^{-0.1\rho}(\pi - |\phi|)}{z^2 + 10} \\ &= 5 \left[\frac{e^{-0.1\rho}(-0.1 - 1)}{(0.1)^2} (-0.101) \right]_0^{10} \times \int_{-\infty}^\infty 2 \int_0^{2\pi} \frac{(\pi - \phi) dz}{z^2 + 10} \end{aligned}$$

$$\begin{aligned} Q &= 5 \times 26.4 \int_{-\infty}^\infty \frac{\pi^2 dz}{z^2 + 10} \\ &= 5 \times 26.4 \times \pi^2 \left[\frac{1}{\sqrt{10}} \tan^{-1} \frac{z}{\sqrt{10}} \right]_{-\infty}^\infty = 129 \text{ mC} \end{aligned}$$

$$\begin{aligned} 13. (A) \quad \phi &= 10 \int_{0.9\pi}^{1.1\pi} \int_0^{25^\circ} \int_4^5 (r-4)(r-5) \sin \theta \cdot \sin \frac{\phi}{2} r^2 \sin \theta \, d\rho d\theta d\phi \\ &= 10 \left[\frac{r^5}{5} - \frac{9r^4}{4} + \frac{20r^3}{3} \right]_4^5 \left[\frac{1}{2} \theta - \frac{1}{4} \sin 2\theta \right]_0^{25^\circ} \times \left[-2 \cos \frac{\phi}{2} \right]_{0.9\pi}^{1.1\pi} \\ &= 10 [-3.39][0.0266][0.626] = 0.57 \text{ C} \end{aligned}$$

$$\begin{aligned} 14. (D) \quad \mathbf{E}_p &= \frac{\rho_L}{2\pi\epsilon_o} \frac{\mathbf{R}_p}{|\mathbf{R}_p|^2}, \\ \mathbf{R}_p &= (1, 2, 3) - (1, -2, 5) = (0, 4, -2) \\ |\mathbf{R}_p|^2 &= 20, \end{aligned}$$

$$\mathbf{E}_p = \frac{5 \times 10^{-9}}{2\pi\epsilon_o} \left[\frac{4\mathbf{u}_y - 2\mathbf{u}_z}{20} \right] = 18\mathbf{u}_y - 9\mathbf{u}_z$$

15. (C) With $z=0$, the general field is

$$\mathbf{E}_{z=0} = \frac{\rho_L}{2\pi\epsilon_o} \left[\frac{(y+2)\mathbf{u}_y - 5\mathbf{u}_z}{(y+2)^2 + 25} \right]$$

we require $|\mathbf{E}_z| = |2\mathbf{E}_y|$

$$\text{So } 2(y+2) = 5 \Rightarrow y = \frac{1}{2}$$

$$\mathbf{E} = \frac{6.25}{2\pi\epsilon_o} \left[\frac{2.5\mathbf{u}_y - 5\mathbf{u}_z}{6.25 + 25} \right] = 9\mathbf{u}_y - 18\mathbf{u}_z$$

$$16. (C) \quad E_p = \frac{\rho_L}{2\pi\epsilon_o} \left[\frac{\mathbf{R}_{+Q}}{|\mathbf{R}_{+Q}|^2} - \frac{\mathbf{R}_{-Q}}{|\mathbf{R}_{-Q}|^2} \right]$$

$$\mathbf{R}_{+Q} = (6, 0, 6) - (0, 3, 6) = (6, -3, 0)$$

$$\mathbf{R}_{-Q} = (6, 0, 6) - (0, -3, 6) = (6, 3, 0)$$

$$\mathbf{E}_p = 20nC \times 2 \times 9 \times 10^9 \left[\frac{6\mathbf{u}_x - 3\mathbf{u}_y}{36 + 9} - \frac{6\mathbf{u}_x + 3\mathbf{u}_y}{36 + 9} \right]$$

$$= -48\mathbf{u}_y \text{ V/m}$$

$$17. (B) \quad \mathbf{E}_p = \frac{\rho_L}{2\pi\epsilon_o} \left[\frac{\mathbf{R}_{xp}}{|\mathbf{R}_{xp}|^3} + \frac{\mathbf{R}_{yp}}{|\mathbf{R}_{yp}|^3} + \frac{\mathbf{R}_{zp}}{|\mathbf{R}_{zp}|^3} \right]$$

$$\mathbf{R}_{xp} = (-3, 0, -1) - (-3, 0, 0) = (0, 2, -1)$$

$$\text{Similarly } \mathbf{R}_{yp} = (-3, 0, -1), \mathbf{R}_{zp} = (-3, 2, 0)$$

$$\mathbf{E}_p = 100 \times 10^{-9} \times 2 \times 9 \times 10^{-9}$$

$$\begin{aligned} \mathbf{E}_p &= 100 \times 10^{-9} \times 2 \times 9 \times 10^{-9} \times \left[\frac{2\mathbf{u}_y - \mathbf{u}_z}{5} + \frac{-3\mathbf{u}_x - \mathbf{u}_z}{10} + \frac{-3\mathbf{u}_x + 2\mathbf{u}_y}{13} \right] \\ &= -0.96\mathbf{u}_x + \mathbf{u}_y - 0.54\mathbf{u}_z \text{ kV/m} \end{aligned}$$

$$18. (C) \text{ At } y=4, \mathbf{E} = \frac{\rho_L}{2\pi\epsilon_o} \mathbf{u}_y,$$

$$d\mathbf{F} = dq\mathbf{E} = \rho_L dz \mathbf{E}$$

$$\mathbf{F} = \int_0^1 \frac{\rho_L^2 dz \mathbf{u}_y}{2\pi\epsilon_o} (6) = 18.75\mathbf{u}_y \mu\text{N}$$

$$19. (A) \quad E = \int \int \frac{\rho_s dS}{4\pi\epsilon_o} \frac{\mathbf{R} - \mathbf{R}'}{|\mathbf{R} - \mathbf{R}'|^3}$$

where $\mathbf{R} = 3\mathbf{u}_x$ and $\mathbf{R}' = y\mathbf{u}_y + z\mathbf{u}_z$,

$$\mathbf{E}_{PA} = \frac{\rho_L}{4\pi\epsilon_o} \int_{-2}^\infty \int_{-2}^2 \frac{3\mathbf{u}_x - y\mathbf{u}_y - z\mathbf{u}_z}{(9 + y^2 + z^2)^{3/2}} dy dz$$

Due to odd function

$$\mathbf{E}_{PA} = \frac{\rho_L}{4\pi\epsilon_o} \int_{-2}^\infty \int_{-2}^2 \frac{3\mathbf{u}_x dy dz}{(9 + y^2 + z^2)^{3/2}}$$

So there is only x component.

20. (D) There will be z component of $\mathbf{E}$ only

$$\mathbf{R} = z\mathbf{u}_z, \mathbf{R}' = \rho\mathbf{u}_\rho, \mathbf{R} - \mathbf{R}' = z\mathbf{u}_z - \rho\mathbf{u}_\rho$$

$$E_{z,Pa} = \frac{\rho_s}{4\pi\epsilon_o} \int_0^{2\pi} \int_0^{0.2} \frac{z\rho d\rho d\phi}{(z^2 + \rho^2)^{3/2}}$$

$$= \frac{2\pi\rho_s z}{4\pi\epsilon_o} \left[\frac{1}{\sqrt{z^2 + \rho^2}} \right]_0^{0.2} = \frac{\rho_s z}{\epsilon_o} \left[\frac{1}{\sqrt{z^2}} - \frac{1}{\sqrt{z^2 + 0.04}} \right]$$

$$E_z = \frac{\rho_s}{\epsilon_o} \left[1 - \frac{z}{\sqrt{z^2 + 0.04}} \right], \text{ at } z=0.5, E_z = 20.2 \text{ V/m}$$

21. (A) Since charge sheet are infinite, the field magnitude associated with each one will be $\rho_s/2\epsilon_o$, which is position independent. The field direction will depend on which side of a given sheet one is positioned.

$$\begin{aligned} E_A &= \left[\frac{10 \times 10^{-9}}{2\epsilon_o} \mathbf{u}_x - \frac{40 \times 10^{-9}}{2\epsilon_o} \mathbf{u}_y - \frac{50 \times 10^{-9}}{2\epsilon_o} \mathbf{u}_z \right] \\ &= 0.56\mathbf{u}_x + 2.23\mathbf{u}_y - 2.8\mathbf{u}_z \end{aligned}$$

$$22. (A) \frac{dy}{dx} = \frac{E_y}{E_x} = \frac{-15x^2y}{5x^3} = \frac{-3y}{x}$$

$$\Rightarrow \frac{dy}{y} = -\frac{3dx}{x} \Rightarrow \ln y = -3 \ln x \ln Cy = \frac{C_1}{x^3},$$

$$\text{At P, } 2 = \frac{C_1}{4^3} \Rightarrow C_1 = 128 \Rightarrow y = \frac{128}{x^3}$$

23. (A) This point lies in the center of a symmetric arrangement of line charges, whose field will all cancel at that point. Thus $\mathbf{D}$ arises from the point charge alone

$$\mathbf{D} = \frac{10 \times 10^{-9} (-3\mathbf{u}_y + 4\mathbf{u}_z)}{4\pi (3^2 + 4^2)^{1.5}},$$

$$\mathbf{D} = -19.1\mathbf{u}_y + 25.5\mathbf{u}_z \text{ pC/m}^2$$

$$24. (C) h_1 = 2\sqrt{4^2 - 1} = 7.75,$$

$$h_2 = 2\sqrt{4^2 - 1} = 6.93$$

$$Q_T = 2 \times 7.75 \times 40\text{n} + 2 \times 6.93 \times 50 + 20\text{n} = 1.33 \text{ } \mu\text{C}$$

$$25. (D) Q = \int_{0.01}^{0.05} \int_{30}^{90} 5e^{-20z} (0.08) d\phi dz \text{ nC}$$

$$= \left(\frac{\pi}{2} - \frac{\pi}{6} \right) (5)(0.08) \left(-\frac{1}{20} \right) e^{-20z} \Big|_{0.01}^{0.05}$$

$$= 9.45 \times 10^{-3} \text{ nC} = 9.45 \text{ pC}$$

26. (A) Out of the 6 surface only 2 will contribute to the net outward flux. The y component of D will penetrate the surface $y=0$ and $y=z$ and net flux will be zero. At $x=0$ plane $D_x=0$ and at $z=0$ plane $D_z=0$.

This leaves the 2 remaining surfaces at $x=2$ and $z=5$.

The net outward flux become

$$f = \int_0^5 \int_0^3 \mathbf{D}|_{x=2} \cdot \mathbf{u}_x dydz + \int_0^3 \int_0^2 \mathbf{D}|_{z=2} \cdot \mathbf{u}_z dzdy$$

$$= 5 \int_0^3 4(2)y dy + 2 \int_0^3 4(5)y dy = 360 \text{ C.}$$

$$27. (D) Q = \int_0^{2\pi} \int_0^\pi \int_0^{0.001} 2e^{-1000r} r^2 \sin \theta dr d\theta d\phi$$

$$Q = 8\pi \left[\frac{-r^2 e^{-1000r}}{1000} \Big|_0^{0.001} + \frac{2e^{-1000r}(-1000r - 1)}{1000(1000)^2} \Big|_0^{0.001} \right]$$

$$Q = 4 \times 10^{-9} \text{ nC,}$$

$$D_r = \frac{Q}{4\pi r^2} = \frac{4.0 \times 10^{-9}}{4\pi \times (0.001)^2} = 0.32 \text{ C/m}^2$$

$$28. (C) 4\pi r^2 D_r = 4\pi (2^2) 20 \times 10^{-9} \text{ C/m}^2,$$

$$D_r = \frac{80}{r^2} \text{ nC/m}^2$$

$$29. (C) Q = \oint_S \mathbf{D} \cdot \mathbf{n} dS = \underbrace{\int_0^a \int_0^a 2ay dy dz}_{\text{Front}} + \underbrace{\int_0^a \int_0^a -2(0) dy dz}_{\text{Back}} \\ + \underbrace{\int_0^a \int_0^a -x^2 dx dz}_{\text{Left}} + \underbrace{\int_0^a \int_0^a x^2 dx dz}_{\text{Right}} + \underbrace{\int_0^a \int_0^a -6(0)^3 dx dy}_{\text{Bottom}} + \underbrace{\int_0^a \int_0^a 6a^3 dx dy}_{\text{Top}} \\ = a^4 + 0 - \frac{a^4}{3} + \frac{a^4}{3} + 0 + 6a^5 = a^4 + 6a^5$$

$$30. (C) \nabla \cdot \mathbf{D} = \frac{\partial D_y}{\partial y} = 20x^4 y^3 z^4$$

Center of cube = (3.05 1.05 2.05) and Volume

$$\nabla V = (0.1)^3 = 0.001$$

$$\phi = 20 (3.05)^4 (1.05)^3 (2.05)^3 (2.05)^4 0.001 = 35.4$$

$$31. (C) \Phi = (\nabla \cdot \mathbf{D}) \Delta v$$

$$= \frac{\partial(x)}{\partial x} \left(\frac{4}{3} \pi (0.003)^3 \right) = 113.1 \text{ nC}$$

$$32. (B) \rho_v = \nabla \cdot \mathbf{D} = \left[\frac{10y}{z} + 0 + \left(\frac{2 + 10x^2y}{z^3} \right) \right]_{(-2, 3, 5)} = 8.96$$

$$33. (C) \nabla \cdot \mathbf{D} = \frac{1}{r^2} \frac{d}{dr} (r^2 \times 2r) = 6,$$

$$\int_V \nabla \cdot \mathbf{D} dv = 6 \times (0.4)^3 = 0.38$$

$$34. (C) dW = -q\mathbf{E} \cdot d\mathbf{L}$$

$$= -5(4\mathbf{u}_x - 3\mathbf{u}_y + 5\mathbf{u}_z) \cdot \frac{(\mathbf{u}_x + \mathbf{u}_y + \mathbf{u}_z)(2)}{\sqrt{3}}$$

$$= -\frac{10}{\sqrt{3}} (4 - 3 + 5) = -34.64 \text{ J}$$

35. (B) The vector in this direction is

$$(2, 1, 4) - (7, 2, 3) = (1, -1, 1)$$

$$\mathbf{u}_{PQ} = \frac{\mathbf{u}_x - \mathbf{u}_y + \mathbf{u}_z}{\sqrt{3}}, dW = -q\mathbf{E} \cdot d\mathbf{L}$$

$$= -(60 \times 10^{-6}) \left(100\mathbf{u}_p \cdot \frac{(\mathbf{u}_x - \mathbf{u}_y + \mathbf{u}_z)(2 \times 10^{-3})}{\sqrt{3}} \right)$$

$$= -12 \times \frac{10^{-6}}{\sqrt{3}} (\mathbf{u}_p \cdot \mathbf{u}_x - \mathbf{u}_p \cdot \mathbf{u}_y)$$

$$\text{At } \rho, \phi = \tan^{-1} \left(\frac{2}{1} \right) = 63.4^\circ$$

$$\mathbf{u}_p \cdot \mathbf{u}_x = \cos 63.4^\circ = 0.447,$$

$$\mathbf{u}_p \cdot \mathbf{u}_y = \sin 63.4^\circ = 0.894$$

$$dW = 3.1 \mu\text{J}$$

$$36. (A) W = -q \int \mathbf{E} \cdot d\mathbf{L}$$

$$\begin{aligned}
&= -10 \int (2x\mathbf{u}_x - 3y^2\mathbf{u}_y + 4\mathbf{u}_z)(dx\mathbf{u}_x + dy\mathbf{u}_y + dz\mathbf{u}_z) \\
&= -10 \int_0^3 2xdx + 10 \int_0^1 3y^2dy - 10 \int_0^1 4dz = -40 \text{ J}
\end{aligned}$$

37. (D) $r_A = 2 \text{ cm}$, $r_B = 3 \text{ cm}$

$$\begin{aligned}
V(r) &= \frac{Q}{4\pi\epsilon_0 r} = \frac{4\pi a^2 \rho_s}{4\pi\epsilon_0 r} = \frac{(6 \times 10^{-3})^2 \times 30 \times 10^{-9}}{8.85 \times 10^{-12} \times r} \\
V(r) &= \frac{0.122}{r}, \quad V_{AB} = V_A - V_B = \frac{0.122}{0.02} - \frac{0.122}{0.03} = 2.03
\end{aligned}$$

38. (D) $W = -\frac{Q_1 Q_2}{4\pi\epsilon_0} \int_{r_A}^{r_B} \frac{dr}{r^2} = \frac{Q_1 Q_2}{4\pi\epsilon_0} \left[\frac{1}{r_B} - \frac{1}{r_A} \right]$

$r_A = r_B$, $W = 0$

39. (B) $V_P(s) = \frac{Q}{4\pi\epsilon_0 r}$,

$$V_l(\rho) = -\int \frac{\rho_L d\rho}{2\pi\epsilon_0 \rho} + C_1 = -\frac{\rho_L}{2\pi\epsilon_0} \ln \rho + C_1$$

$$V_s(z) = -\int \frac{\rho_s}{2\epsilon_0} dz + C_2 = \frac{\rho_s z}{2\epsilon_0} + C_2$$

$$V = \frac{Q}{4\pi\epsilon_0 r} - \frac{\rho_L}{2\pi\epsilon_0} \ln \rho - \frac{\rho_s}{2\pi\epsilon_0} z + C$$

Here r, ρ, z are the scalar distance from the charge.

$$r = \sqrt{2^2 + 5^2} = \sqrt{29}, \quad \rho = \sqrt{(5-4)^2} = 1, \quad z = 5$$

By putting these value. $C = -1.93 \times 10^3$

$$\text{At point N, } r = \sqrt{(2-1)^2 + 2^2 + z^2} = \sqrt{14}$$

$$\rho = \sqrt{1^2 + 1^2} = \sqrt{2}, \quad z = 3, \quad V_B = 1.98 \text{ kV}$$

40. (D) $V_\rho = \int_{-\infty}^{\infty} \frac{\rho_L dz}{4\pi\epsilon_0 R} = \int_{-\infty}^{\infty} \frac{6 \times 10^{-9} dz}{4\pi\epsilon_0 (z^2 + 1)^{3/2}} = 108 \text{ V}$

41. (A) $V_\rho = \iint \frac{\rho_s dS}{4\pi\epsilon_0 |\mathbf{R} - \mathbf{R}'|}$,

$$\mathbf{R} = z\mathbf{u}_z, \mathbf{R}' = \rho\mathbf{u}_\rho, \quad dS = \rho d\rho d\phi$$

$$V_\rho = \int_0^{2\pi} \int_{0.01}^{0.03} \frac{(5 \times 10^{-9}) \rho^2 d\rho d\phi}{4\pi\epsilon_0 \sqrt{\rho^2 + z^2}},$$

$$V_\rho = \frac{5 \times 10^{-9}}{2\epsilon_0} \left[\frac{\rho}{2} \sqrt{\rho^2 + z^2} - \frac{z^2}{2} \ln(\rho + \sqrt{\rho^2 + z^2}) \right]_{0.01}^{0.03}$$

At $z = 0.02$, $V_\rho = 0.081 \text{ V}$

42. (C) $\mathbf{E} = -\nabla V$

$$\begin{aligned}
&= -\left[2y^2 z^2 + \frac{6x}{x^2 + 2y^2 + 3z^2} \right] \mathbf{u}_x \\
&\quad - \left[4xyz^3 + \frac{12y}{x^2 + 2y^2 + 3z^2} \right] \mathbf{u}_y
\end{aligned}$$

$$-\left[6xy^2 z^2 + \frac{18z}{x^2 + 2y^2 + 3z^2} \right] \mathbf{u}_z$$

$\mathbf{E}_p = 7.1\mathbf{u}_x + 22.8\mathbf{u}_y - 71.1\mathbf{u}_z \text{ V/m}$,

$|\mathbf{E}| = 75 \text{ V/m}$

43. (D) $\mathbf{E} = -\nabla V = -\frac{dV}{dr} \mathbf{u}_r = -(0.6)(70)r^{-0.4} \mathbf{u}_r$

$$= -42r^{-0.4} \mathbf{u}_r \text{ V/m},$$

$$\mathbf{D} = \epsilon_0 \mathbf{E} = -42r^{-0.4} \epsilon_0 \mathbf{u}_r \text{ C/m}^2$$

$$\rho_v = \nabla \cdot \mathbf{D} = \frac{1}{r^2} \frac{d}{dr} (r^2 D_r) = \frac{1}{r^2} \frac{d}{dr} (-42\epsilon_0 r^{1.6}) = -67.2 \frac{\epsilon_0}{r^{1.4}}$$

At $r = 0.6 \text{ m}$,

$$\rho_v = -\frac{67.2 \times 8.85 \times 10^{-12}}{(0.6)^{1.4}} = -1.22 \text{ nC/m}^3$$

44. (A) $\mathbf{E} = -\nabla V = -\frac{\partial V}{\partial r} \mathbf{u}_r - \frac{1}{r} \frac{\partial V}{\partial \theta} \mathbf{u}_\theta$

$$= -160r \cos \theta \mathbf{u}_r + 80r \sin \theta \mathbf{u}_\theta \text{ V/m}$$

$$\mathbf{D} = \epsilon_0 \mathbf{E} = -80\epsilon_0 (2r \cos \theta \mathbf{u}_r - r \sin \theta \mathbf{u}_\theta)$$

$$\rho_v = \nabla \cdot \mathbf{D} = \left[\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 D_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (D_r \sin \theta) \right]$$

$$\rho_v = -80\epsilon_0 \left[\frac{1}{r} 2 \times 3r^2 \cos \theta - \frac{12 \sin \theta \cos \theta}{r \sin \theta} \right]$$

$$\rho_v = -320\epsilon_0 \cos \theta = -2.45 \text{ nC/m}^3$$

45. (C) $\mathbf{E} = -\nabla V = -\frac{\partial V}{\partial \rho} \mathbf{u}_\rho - \frac{1}{\rho} \frac{\partial V}{\partial \phi} \mathbf{u}_\phi$

$$= -(50 + 150 \sin \phi) \mathbf{u}_\rho - (150 \cos \phi) \mathbf{u}_\phi,$$

$$\mathbf{D} = \epsilon_0 \mathbf{E}, \quad \rho_v = \nabla \cdot \mathbf{D} = \epsilon_0 \nabla \cdot \mathbf{E}$$

$$\rho_v = \epsilon_0 \left[\frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho E_\rho) + \frac{1}{\rho} \frac{\partial}{\partial \phi} E_\phi \right]$$

$$= \epsilon_0 \left[-\frac{(50 + 150 \sin \phi)}{\rho} + \frac{150 \sin \phi}{\rho} \right] = -\frac{50\epsilon_0}{\rho} \text{ C/m}^2$$

$$Q = \int_0^1 \int_0^{2\pi} \int_0^2 \frac{50\epsilon_0}{\rho} \rho d\rho d\phi dz = -2\pi(50)\epsilon_0 2 = -5.56 \text{ nC}$$

46. (A) $V = \frac{Qd \cos \theta}{4\pi\epsilon_0 r^2} = \frac{100 \cos \theta}{r^2}$,

$$\mathbf{E} = -\nabla V = -\left(\frac{\partial V}{\partial r} \mathbf{u}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \mathbf{u}_\theta \right)$$

$$= \frac{100}{r^3} (2 \cos \theta \mathbf{u}_r + \sin \theta \mathbf{u}_\theta),$$

$$|\mathbf{E}| = 100(4 \cos^2 \theta + \sin^2 \theta)^{1/2} = 100 \times \sqrt{\frac{5}{2}} = 158.11 \text{ V/m}$$

47. (A) $\mathbf{E} = \frac{2 \times 10^{-9}}{4\pi\epsilon_0 r^3} [2 \cos \theta \mathbf{u}_r + \sin \theta \mathbf{u}_\theta]$,

$y = z$ lies at $\theta = 45^\circ$

$$E_0 = \frac{2 \times 10^{-9}}{4\pi\epsilon_0 r^3} \frac{1}{\sqrt{2}} = 10^{-3} \quad (\text{required})$$

$$r^3 = 12.73 \times 10^3, \quad r = 23.35$$

$$48. (A) V = \frac{\mathbf{P} \cdot (\mathbf{R} - \mathbf{R}')}{4\pi\epsilon_0 |\mathbf{R} - \mathbf{R}'|^3}$$

where $\mathbf{R} - \mathbf{R}' = \mathbf{Q} - \mathbf{P} = (1, 1, 8)$

$$\text{So } V_P = \frac{(3\mathbf{u}_x - 5\mathbf{u}_y + 10\mathbf{u}_z) \cdot (\mathbf{u}_x + \mathbf{u}_y + 8\mathbf{u}_z) \times 10^{-9}}{4\pi\epsilon_0 (1 + 1 + 8^2)^{1.5}}$$

$$= 1.31 \text{ V}$$

$$49. (A) \mathbf{E} = -\nabla V = 40 \left[\frac{1}{x^2 y z} \mathbf{u}_x + \frac{1}{x y^2 z} \mathbf{u}_y + \frac{1}{x y z^2} \mathbf{u}_z \right]$$

$$W_e = \frac{\epsilon_0}{2} \iiint \mathbf{E} \cdot \mathbf{E} dv$$

$$W_e = 800\epsilon_0 \int_1^2 \int_1^2 \int_1^2 \left[\frac{1}{x^4 y^2 z^2} + \frac{1}{x^2 y^4 z^2} + \frac{1}{x^2 y^2 z^4} \right] dx dy dz$$

$$= 1548 \text{ pJ}$$

$$50. (A) W = \frac{1}{2} \sum_{n=1}^4 q_n V_n$$

$$V_1 = V_{21} + V_{31} + V_{41} = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{0.04} + \frac{1}{0.04} + \frac{1}{0.04\sqrt{2}} \right]$$

$$V_1 = V_2 = V_3 = V_4$$

$$W = \frac{1}{2} (4) q_1 V_1 = \frac{2 \times (1.2 \times 10^{-9})^2}{4\pi\epsilon_0 (0.04)} \left[2 + \frac{1}{\sqrt{2}} \right] = 1.75 \text{ } \mu\text{J}$$

$$51. (B) I = \iint_S \mathbf{J} \cdot \mathbf{n} \big|_{y=1} dS = \iint_{0 \leq x \leq 1, 0 \leq z \leq 1} \mathbf{J} \cdot \mathbf{u}_y \big|_{y=1} dx dz$$

$$= \int_0^2 \int_0^1 10^5 \cos(2x) 2^{-2y} dx dz = 12307 \text{ kA} = 12.3 \text{ MA}$$

$$52. (C) I = \iint_S \mathbf{J} \cdot \mathbf{n} dS$$

$$= \int_0^{2\pi} \int_{0.1\pi}^{0.3\pi} \frac{800 \sin \theta}{(0.80)^2 + 4} (0.80)^2 \sin \theta d\theta d\phi = 154.8 \text{ A}$$

$$53. (D) F = ma = qE,$$

$$a = \frac{qE}{m} = \frac{(-1.602 \times 10^{-19})(-4 \times 10^6)}{9.11 \times 10^{-31}} \mathbf{u}_z = 7.0 \times 10^{17} \mathbf{u}_z \text{ m/s}^2,$$

$$v = at = 7.0 \times 10^{17} t \mathbf{u}_z \text{ m/s}$$

$$54. (D) \frac{\partial \rho_V}{\partial t} = -\nabla \cdot \mathbf{J} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho J_\rho) + \frac{\partial J_z}{\partial z}$$

$$= \frac{1}{\rho} \frac{\partial}{\partial \rho} (25) + \frac{\partial}{\partial z} \left(\frac{-20}{\rho^2 + 0.01} \right) = 0$$

$$55. (A) I = \iint_S \mathbf{J} \cdot \mathbf{n} \big|_{z=0.2} dS$$

$$= \int_0^{2\pi} \int_0^{0.4} \frac{40}{\rho^2 + (0.1)^2} \rho d\rho d\phi = \frac{40}{2} \log[\rho^2 + (0.1)^2] \bigg|_0^{0.4} (2\pi)$$

$$= 40\pi \log 17 = 356 \text{ A}$$

$$56. (B) \text{ So } E_{al} = E_{st} = \frac{J_{al}}{\sigma_{al}} = \frac{J_{st}}{\sigma_{st}} \Rightarrow J_{al} = \frac{\sigma_{ac}}{\sigma_{st}} J_{st}$$

$$I = \pi(2 \times 10^{-3}) J_{st} + \pi[(4 \times 10^{-3})^2 - (2 \times 10^{-3})^2] J_{al} = 80$$

$$\text{Solving } J_{st} = 3.2 \times 10^5 \text{ A/m}^2$$

$$57. (B) J = \frac{4}{2\pi\rho l} \mathbf{u}_\rho \text{ A/m}^2,$$

$$E = \frac{J}{\sigma} = \frac{4}{2\pi\rho l \sigma} = \frac{12.73}{\rho l} \mathbf{u}_\rho \text{ V/m}$$

$$V = -\int_5^3 \mathbf{E} \cdot d\mathbf{L} = \int_3^5 \frac{12.73}{\rho l} \mathbf{u}_\rho \cdot \mathbf{u}_\rho d\rho = \frac{12.73}{l} \ln \frac{5}{3} = \frac{6.51}{l} \text{ V}$$

$$R = \frac{V}{I} = \frac{6.51}{4l} = \frac{1.63}{\rho} \Omega$$

$$58. (D) \mathbf{E} = -\nabla V = -\left(\frac{\partial V}{\partial \rho} \mathbf{u}_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \mathbf{u}_\phi + \frac{\partial V}{\partial z} \mathbf{u}_z \right)$$

$$= -z^2 \cos \phi \mathbf{u}_\rho + \frac{(\rho + 1)}{\rho} z^2 \sin \phi \mathbf{u}_\phi - 2(\rho + 1)z \cos \phi \mathbf{u}_z$$

$$\mathbf{E} = -1.82 \mathbf{u}_\phi + 14.5 \mathbf{u}_\rho - 2.67 \mathbf{u}_z \text{ V/m}$$

$$\rho_s = \epsilon_0 \mathbf{E} \cdot \mathbf{n} \big|_s = \epsilon_0 \frac{\mathbf{E} \cdot \mathbf{E}}{|\mathbf{E}|}$$

$$\rho_s = \epsilon_0 \sqrt{1.82^2 + 14.5^2 + 2.67^2} = 131.5 \text{ pC/m}^2$$

$$59. (C) V = \frac{40 \cos \frac{\pi}{3} \sin \frac{\pi}{2}}{2^3} = 2.5 \text{ V}$$

So the equation of the surface is

$$\frac{40 \cos \theta \sin \phi}{r^2} = 2.5, \quad 16 \cos \theta \sin \phi = r^3$$

$$60. (A) \mathbf{E} = -\nabla V, \quad \mathbf{D} = \epsilon \mathbf{E} = -\epsilon_0 \nabla V$$

$$= -\epsilon_0 \left[200z \frac{\partial}{\partial x} \frac{x}{(x^2 + 4)} \mathbf{u}_x + \frac{200x}{x^2 + 4} \mathbf{u}_z \right] \text{ C/m}^2$$

$$\mathbf{D}_{(z=0)} = -\frac{200\epsilon_0 x}{x^2 + 4} \mathbf{u}_z \text{ C/m}^2,$$

$$\rho_s = \mathbf{D} \cdot \mathbf{u}_z \big|_{z=0} = \frac{-200\epsilon_0 x}{x^2 + 4} \text{ C/m}^2,$$

$$Q = \int_{-3}^0 \int_0^2 \frac{-200\epsilon_0 x}{x^2 + 4} dx dy = -(3)(200)\epsilon_0 \frac{1}{2} \ln[x^2 + 4] \bigg|_0^2$$

$$= -300 \ln 2 = -1.84 \text{ nC}$$

61. (B) $\mathbf{E} = -\nabla V$, $-3xy^2z^3$ satisfy this equation.

62. (D) The plane can be replaced by -60 nC at $Q(2, 5, -6)$.

$$\mathbf{R} = (5, 3, 1) - (2, 4, 6) = (3, -1, -5)$$

$$\mathbf{R}' = (5, 3, 1) - (2, 4, -6) = (3, -1, 7),$$

$$|\mathbf{R}| = \sqrt{35}, |\mathbf{R}'| = \sqrt{59}$$

$$V_p = \frac{q}{4\pi\epsilon_0 R} - \frac{q}{4\pi\epsilon_0 R'} = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{\sqrt{35}} - \frac{1}{\sqrt{59}} \right]$$

$$V_p = 60 \times 10^{-9} \times 9 \times 10^{-9} \left[\frac{1}{\sqrt{35}} - \frac{1}{\sqrt{59}} \right] = 21 \text{ V}$$

63. (A) Using method of images

$$V_p - V_0 = -\frac{\rho_L}{2\pi\epsilon_0} \ln \left[\frac{\text{Final Distance from the charge}}{\text{Initial Distance from the charge}} \right]$$

$$V_0 = 0,$$

$$V_p = -\frac{\rho_L}{2\pi\epsilon_0} \left[\ln \frac{1}{2} + \ln \frac{\sqrt{1^2 + (2-1)^2}}{1} - \ln \frac{\sqrt{1^2 + (2-(-2))^2}}{1} - \ln \frac{\sqrt{1^2 + 3^2}}{1} \right]$$

$$= 2.40 \text{ kV}$$

$$64. (C) \mathbf{P} = \mathbf{D} - \epsilon_0 \mathbf{E} = \mathbf{D} - \frac{\mathbf{D}}{\epsilon_r} = \frac{\mathbf{D}}{\epsilon_r} (\epsilon_r - 1)$$

$$\mathbf{P} = (2\mathbf{u}_x - 4\mathbf{u}_y + 5\mathbf{u}_z) \frac{(2.4 - 1) \times 10^{-9}}{2.4}$$

$$\mathbf{P} = 12\mathbf{u}_x - 2.3\mathbf{u}_y + 2.9\mathbf{u}_z \text{ nC/m}^2$$

$$65. (C) \mathbf{D}_{n1} = \mathbf{D}_{n2} \Rightarrow \epsilon_1 \mathbf{E}_{N1} = \epsilon_2 \mathbf{E}_{N2}$$

$$= 2 \times 1.5\epsilon_0 = 2.5\epsilon_0 E_{n2} \Rightarrow E_{n2} = 12$$

$$E_{t1} = E_{t2}, \quad \mathbf{E}_2 = 1.2\mathbf{u}_x - 3\mathbf{u}_y + 1\mathbf{u}_z$$

66. (D) The unit vector that is normal to the surface is

$$\mathbf{u}_N = \frac{\nabla F}{|\nabla F|} = \frac{\mathbf{u}_x - \mathbf{u}_y + 2\mathbf{u}_z}{\sqrt{6}},$$

$$E_{n1} = \mathbf{E}_1 \cdot \mathbf{u}_N = \frac{1}{\sqrt{6}} [100 - 200 - 100] = -81.7 \text{ V/m}$$

$$\mathbf{E}_{n1} = -81.7 \frac{1}{\sqrt{6}} [\mathbf{u}_x - \mathbf{u}_y + 2\mathbf{u}_z]$$

$$= -33.33\mathbf{u}_x + 33.33\mathbf{u}_y - 66.67\mathbf{u}_z \text{ V/m}$$

$$\mathbf{E}_{t1} = \mathbf{E}_1 - \mathbf{E}_{n1} = 133.3\mathbf{u}_x + 166.7\mathbf{u}_z + 16.67\mathbf{u}_z$$

$$\mathbf{E}_{t1} = \mathbf{E}_{t2} \text{ and } \mathbf{D}_{n1} = \mathbf{D}_{n2}$$

$$\Rightarrow \epsilon_{r1} \epsilon_0 \mathbf{E}_{n1} = \epsilon_{r2} \epsilon_0 \mathbf{E}_{n2}$$

$$\mathbf{E}_{n2} = \frac{\epsilon_{r1}}{\epsilon_{r2}} \mathbf{E}_{n1} = \frac{1}{4} \mathbf{E}_{n1}, \quad \mathbf{E}_2 = \mathbf{E}_{t2} + \frac{1}{4} \mathbf{E}_{n1}$$

$$= 133.3\mathbf{u}_x + 166.7\mathbf{u}_y + 16.67\mathbf{u}_z - 8.33\mathbf{u}_x + 8.33\mathbf{u}_y - 16.67\mathbf{u}_z$$

$$= 125\mathbf{u}_x + 175\mathbf{u}_y \text{ V/m}$$

$$67. (B) \mathbf{D}_{n1} = \mathbf{D}_{n2}, \quad \epsilon_{r1} \epsilon_0 \mathbf{E}_{n1} = \epsilon_{r2} \epsilon_0 \mathbf{E}_{n2}$$

Since $\mathbf{E}$ is normal to the surface

$$\mathbf{E}_{n2} = \frac{\epsilon_{r1}}{\epsilon_{r2}} \mathbf{E}_{n1} = \frac{2}{5} \times \frac{1000}{r^2} \mathbf{u}_r = \frac{400}{r^2} \mathbf{u}_r \text{ V/m}$$

$$68. (D) \mathbf{D}_{n1} = \mathbf{D}_{n2} \text{ and } \mathbf{E}_{t1} = \mathbf{E}_{t2}, \quad \mathbf{D} = \epsilon \mathbf{E}$$

$$\mathbf{D}_{n1} = 80 \times 3\epsilon_0 = \mathbf{D}_{n2} = 5\epsilon_0 \mathbf{E}_{n1}, \quad \mathbf{E}_{n1} = 48$$

$$\mathbf{E}_2 = 48\mathbf{u}_x - 60\mathbf{u}_y - 40\mathbf{u}_z$$

$$69. (A) \mathbf{D} = \epsilon \mathbf{E} = -f(x) \nabla V,$$

$$\nabla \cdot \mathbf{D} = \rho_v = 0 = \nabla \cdot (-f(x) \nabla V)$$

$$0 = \nabla \cdot (-f(x) \nabla V)$$

$$= - \left[\frac{dF}{dx} \frac{\partial V}{\partial x} + f(x) \frac{\partial^2 V}{\partial x^2} + f(x) \frac{\partial^2 V}{\partial y^2} + f(x) \frac{\partial^2 V}{\partial z^2} \right]$$

$$= - \left[\frac{dF}{dx} \frac{\partial V}{\partial x} + f(x) \nabla^2 V \right]$$

$$\Rightarrow \nabla^2 V = - \frac{1}{f(x)} \frac{dF}{dx} \frac{\partial V}{\partial x}$$

$$70. (A) \rho_v = 0 \Rightarrow \nabla^2 V = 0,$$

$$\nabla^2 V = 16e^{2x} + \frac{\partial^2 f}{\partial x^2} - 6 = 0$$

$$\frac{\partial f}{\partial x} = -16e^{2x} + 6 \Rightarrow \frac{\partial f}{\partial x} = -8e^{2x} + 6x + C_1$$

$$E_x = \frac{\partial V}{\partial x} = 8e^{2x} + \frac{\partial f}{\partial x}$$

$$E_x(0) = 8 + \frac{\partial f}{\partial x} \Big|_{x=0} = 0 \Rightarrow \frac{\partial f}{\partial x} \Big|_{x=0} = -8$$

It follow that $C_1 = 0$

Integrating again

$$f(x_1) = -4e^{2x} + 3x^2 + C, \quad f(0, 1) = -4 + C_2$$

$$V(0, 0) = 0 = 4 + f(0) \Rightarrow C_2 = 0. \quad f(0) = -4$$

$$f(x) = -4e^{2x} + 3x^2,$$

$$V(x, y) = 4e^{2x} - 4e^{2x} + 3x^2 - 3y^2 = 3(x^2 - y^2)$$

Statement for Q.9–11:

An infinite filament on the z -axis carries 10 mA in the $\mathbf{u}_z$ direction. Three uniform cylindrical current sheets are also present at 400 mA/m at $\rho = 1$ cm, -250 mA/m at $\rho = 2$ cm and 300 mA/m at $\rho = 3$ cm.

9. The magnetic field H_ϕ at $\rho = 0.5$ cm is
(A) 0.32 A/m (B) 0.64 A/m

- (C) 1.36 mA/m (D) 0

10. The magnetic field H_ϕ at $\rho = 1.5$ cm is
(A) 1.63 A/m (B) 0.37 A/m

- (C) 2.64 A/m (D) 0

11. The magnetic field H_ϕ at $\rho = 3.5$ cm is
(A) 0.14 A/m (B) 0.56 A/m

- (C) 0.27 A/m (D) 0.96 A/m

Statement for Q.12–14:

In the fig. P8.3.12–14 The region $0 \leq z \leq 2$ is filled with an infinite slab of magnetic material ($\mu_r = 2.5$). The surface of the slab at $z = 0$ and $z = 2$, respectively, carry surface current $30\mathbf{u}_x$ A/m and $-40\mathbf{u}_x$ as shown in fig.

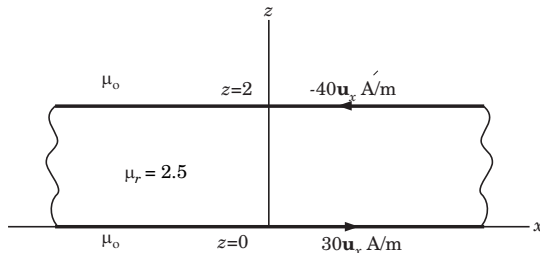


Fig. P8.3.12–14

12. In the region $0 < z < 2$ the $\mathbf{H}$ is
(A) $-35\mathbf{u}_y$ A/m (B) $35\mathbf{u}_y$ A/m

- (C) $-5\mathbf{u}_y$ A/m (D) $5\mathbf{u}_y$ A/m

13. In the region $z < 0$ the $\mathbf{H}$ is
(A) $5\mathbf{u}_y$ A/m (B) $-5\mathbf{u}_y$ A/m

- (C) $10\mathbf{u}_y$ A/m (D) $-10\mathbf{u}_y$ A/m

14. In the region $z > 2$ the $\mathbf{H}$ is
(A) $5\mathbf{u}_y$ A/m (B) $-5\mathbf{u}_y$ A/m

- (C) $35\mathbf{u}_y$ A/m (D) $-35\mathbf{u}_y$ A/m

Statement for Q.15–16:

In the cylindrical region

$$H_\phi = \frac{2}{\rho} + \frac{\rho}{2} \text{ for } \rho \leq 0.6$$

$$H_\phi = \frac{3}{\rho} \text{ for } \rho > 0.6$$

15. The current density $\mathbf{J}$ for $\rho < 0.6$ mm is
(A) $2\mathbf{u}_z$ A/m (B) $-2\mathbf{u}_z$ A/m

- (C) $\mathbf{u}_z$ A/m (D) 0

16. The current density $\mathbf{J}$ for $\rho > 0.6$ mm is
(A) $2\mathbf{u}_z$ A/m (B) $-3\mathbf{u}_z$ A/m

- (C) $3\mathbf{u}_z$ A/m (D) 0

17. An electron with velocity $\mathbf{v} = (3\mathbf{u}_x + 12\mathbf{u}_y - 4\mathbf{u}_z) \times 10^5$ m/s experiences no net forces at a point in a magnetic field $\mathbf{B} = \mathbf{u}_x + 2\mathbf{u}_y + 3\mathbf{u}_z$ mWb/m². The electric field $\mathbf{E}$ at that point is

- (A) $-4.4\mathbf{u}_x + 1.3\mathbf{u}_y + 0.6\mathbf{u}_z$ kV/m

- (B) $4.4\mathbf{u}_x - 1.3\mathbf{u}_y - 0.6\mathbf{u}_z$ kV/m

- (C) $-4.4\mathbf{u}_x + 1.3\mathbf{u}_y + 0.6\mathbf{u}_z$ kV/m

- (D) $4.4\mathbf{u}_x - 1.3\mathbf{u}_y - 0.6\mathbf{u}_z$ kV/m

18. A point charge of 2×10^{-16} C and 5×10^{-26} kg is moving in the combined fields $\mathbf{B} = -3\mathbf{u}_x + 2\mathbf{u}_y - \mathbf{u}_z$ mT and $\mathbf{E} = 100\mathbf{u}_x - 200\mathbf{u}_y + 300\mathbf{u}_z$ V/m. If the charge velocity at $t = 0$ is $\mathbf{v}(0) = (2\mathbf{u}_x - 3\mathbf{u}_y - 4\mathbf{u}_z) 10^5$ m/s, the acceleration of charge at $t = 0$ is

- (A) $600[3\mathbf{u}_x + 2\mathbf{u}_y - 3\mathbf{u}_z] 10^9$ m/s²

- (B) $400[6\mathbf{u}_x + 6\mathbf{u}_y - 3\mathbf{u}_z] 10^9$ m/s²

- (C) $400[6\mathbf{u}_x - 6\mathbf{u}_y + 3\mathbf{u}_z] 10^9$ m/s²

- (D) $800[6\mathbf{u}_x + 6\mathbf{u}_y - \mathbf{u}_z] 10^9$ m/s²

19. An electron is moving at velocity $\mathbf{v} = 4.5 \times 10^7 \mathbf{u}_y$ m/s along the negative y -axis. At the origin, it encounters the uniform magnetic field $\mathbf{B} = 2.5\mathbf{u}_z$ mT, and remains in it up to $y = 2.5$ cm. If we assume that the electron remains on the y -axis while it is in the magnetic field, at $y = 50$ cm the x and z coordinate are respectively

- (A) 1.23 m, 0.23 m (B) -1.23 m, -0.23 m

- (C) -11.7 cm, 0 (D) 11.7 cm, 0

Statement for Q.20–22:

A rectangular loop of wire in free space joins points A(1, 0, 1) to B(3, 0, 1) to C(3, 0, 4) to D(1, 0, 4) to A. The wire carries a current of 6 mA flowing in the $\mathbf{u}_z$ direction from B to C. A filamentary current of 15 A flows along the entire z , axis in the $\mathbf{u}_z$ directions.

20. The force on side BC is

- (A) $-18\mathbf{u}_x$ nN (B) $18\mathbf{u}_x$ nN
(C) $3.6\mathbf{u}_x$ nN (D) $-3.6\mathbf{u}_x$ nN

21. The force on side AB is

- (A) $23.4\mathbf{u}_z$ μ N (B) $16.4\mathbf{u}_z$ μ N
(C) $19.8\mathbf{u}_z$ nN (D) $26.3\mathbf{u}_z$ nN

22. The total force on the loop is

- (A) $36\mathbf{u}_x$ nN (B) $-36\mathbf{u}_x$ nN
(C) $54\mathbf{u}_x$ nN (D) $-54\mathbf{u}_x$ nN

23. Consider the rectangular loop on $z=0$ plane shown in fig. P8.3.23. The magnetic flux density is $\mathbf{B} = 6x\mathbf{u}_x - 9y\mathbf{u}_y + 3z\mathbf{u}_z$ Wb/m². The total force experienced by the rectangular loop is

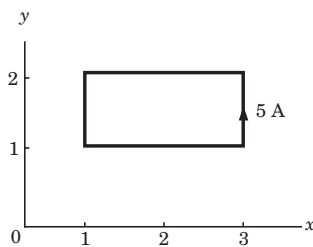


Fig. P8.3.23

- (A) $30\mathbf{u}_z$ N (B) $-30\mathbf{u}_z$ N
(C) $36\mathbf{u}_z$ N (D) $-36\mathbf{u}_z$ N

Statement for Q.24–25:

Three uniform current sheets are located in free space as follows: $8\mathbf{u}_z$ A/m at $y=0$, $-4\mathbf{u}_z$ A/m at $y=1$ and $-4\mathbf{u}_z$ A/m at $y=-1$. Let $\mathbf{F}$ be the vector force per meter length exerted on a current filament carrying 7 mA in the $\mathbf{u}_L$ direction.

24. If the current filament is located at $x=0$, $y=0.5$ and $\mathbf{u}_L = \mathbf{u}_z$, then $\mathbf{F}$ is

- (A) $35.2\mathbf{u}_y$ nN/m (B) $-35.2\mathbf{u}_y$ nN/m
(C) $105.6\mathbf{u}_y$ nN/m (D) 0

25. If the current filament is located at $y=0.5$, $z=0$, and $\mathbf{u}_L = \mathbf{u}_x$, then $\mathbf{F}$ is

- (A) $35.2\mathbf{u}_y$ nN/m (B) $68.3\mathbf{u}_x$ nN/m
(C) $105.6\mathbf{u}_z$ nN/m (D) 0

26. Two infinitely long parallel filaments each carry 100 A in the $\mathbf{u}_z$ direction. If the filaments lie in the plane $y=0$ at $x=0$ and $x=5$ mm, the force on the filament passing through the origin is

- (A) $0.4\mathbf{u}_x$ N/m (B) $-0.4\mathbf{u}_x$ N/m
(C) $4\mathbf{u}_x$ mN/m (D) $-4\mathbf{u}_x$ mN/m

Statement for Q.27–28:

A conducting current strip carrying $\mathbf{K} = 6\mathbf{u}_z$ A/m lies in the $x=0$ plane between $y=0.5$ and $y=1.5$ m. There is also a current filament of $I=5$ A in the $\mathbf{u}_z$ direction on the z -axis.

27. The force exerted on the filament by the current strip is

- (A) $12.2\mathbf{u}_y$ μ N/m (B) $6.6\mathbf{u}_y$ μ N/m
(C) $-12.2\mathbf{u}_y$ μ N/m (D) $-6.6\mathbf{u}_y$ μ N/m

28. The force exerted on the strip by the filament is

- (A) $-6.6\mathbf{u}_y$ μ N/m (B) $6.6\mathbf{u}_y$ μ N/m
(C) $2.4\mathbf{u}_x$ μ N/m (D) $-2.4\mathbf{u}_x$ μ N/m

Statement for Q.29–32:

In a certain material for which $\mu_r = 6.5$,

$$\mathbf{H} = 10\mathbf{u}_x + 25\mathbf{u}_y - 40\mathbf{u}_z \text{ A/m}$$

29. The magnetic susceptibility χ_m of the material is

- (A) 5.5 (B) 6.5
(C) 7.5 (D) None of the above

30. The magnetic flux density $\mathbf{B}$ is

- (A) $82\mathbf{u}_x + 204\mathbf{u}_y - 327\mathbf{u}_z$ μ Wb/m²
(B) $82\mathbf{u}_x + 204\mathbf{u}_y - 327\mathbf{u}_z$ μ A/m
(C) $82\mathbf{u}_x + 204\mathbf{u}_y - 327\mathbf{u}_z$ mT
(D) $82\mathbf{u}_x + 204\mathbf{u}_y - 327\mathbf{u}_z$ mA/m

31. The magnetization $\mathbf{M}$ is

- (A) $75\mathbf{u}_x + 187.5\mathbf{u}_y - 300\mathbf{u}_z$ A/m²
(B) $75\mathbf{u}_x + 187.5\mathbf{u}_y - 300\mathbf{u}_z$ A/m²
(C) $55\mathbf{u}_x + 137.5\mathbf{u}_y - 220\mathbf{u}_z$ A/m²
(D) $55\mathbf{u}_x + 137.5\mathbf{u}_y - 220\mathbf{u}_z$ A/m²

32. The magnetic energy density is

- (A) 19 mJ/m^2 (B) 9.5 mJ/m^2
(C) 16.3 mJ/m^2 (D) 32.6 mJ/m^2

Statement for Q.33–34:

For a given material magnetic susceptibility $\chi_m = 3.1$ and within which $\mathbf{B} = 0.4y\mathbf{u}_z \text{ T}$.

33. The magnetic field $\mathbf{H}$ is

- (A) $986.8y\mathbf{u}_z \text{ kA/m}$ (B) $151.6y\mathbf{u}_z \text{ kA/m}$
(C) $102.7y\mathbf{u}_z \text{ kA/m}$ (D) $77.6y\mathbf{u}_z \text{ kA/m}$

34. The magnetization $\mathbf{M}$ is

- (A) $241y\mathbf{u}_z \text{ kA/m}$ (B) $318.2y\mathbf{u}_z \text{ kA/m}$
(C) $163y\mathbf{u}_z \text{ kA/m}$ (D) None of the above

35. In a material the magnetic field intensity is $H = 1200 \text{ A/m}$ when $B = 2 \text{ Wb/m}^2$. When H is reduced to 400 A/m , $B = 1.4 \text{ Wb/m}^2$. The change in the magnetization $\mathbf{M}$ is

- (A) 164 kA/m (B) 326 kA/m
(C) 476 kA/m (D) 238 kA/m

36. A particular material has $2.7 \times 10^{29} \text{ atoms/m}^3$ and each atom has a dipole moment of $2.6 \times 10^{-30} \mathbf{u}_y \text{ A} \cdot \text{m}^2$. The $\mathbf{H}$ in material is ($\mu_r = 4.2$)

- (A) $2.94\mathbf{u}_y \text{ A/m}$ (B) $0.22\mathbf{u}_y \text{ A/m}$
(C) $0.17\mathbf{u}_y \text{ A/m}$ (D) $2.24\mathbf{u}_y \text{ A/m}$

37. In a material magnetic flux density is 0.02 Wb/m^2 and the magnetic susceptibility is 0.003 . The magnitude of the magnetization is

- (A) 47.6 A/m (B) 23.4 A/m
(C) 16.3 A/m (D) 8.4 A/m

38. A uniform field $\mathbf{H} = -600\mathbf{u}_y \text{ A/m}$ exist in free space. The total energy stored in spherical region 1 cm in radius centered at the origin in free space is

- (A) 0.226 J/m^3 (B) 1.452 J/m^3
(C) 1.68 J/m^3 (D) 0.84 J/m^3

39. The magnetization curve for an iron alloy is approximately given by

$$B = \frac{1}{3} H + H^2 \mu\text{Wb/m}^2$$

If H increases from 0 to 210 A/m , the energy stored per unit volume in the alloy is

- (A) 6.2 MJ/m^3 (B) 1.3 MJ/m^3
(C) 2.3 kJ/m^3 (D) 2.9 kJ/m^3

40. If magnetization is given by $\mathbf{H} = \frac{6}{a}(-y\mathbf{u}_x + x\mathbf{u}_y)$ in a cube of size a , the magnetization volume current density is

- (A) $\frac{12}{a} \mathbf{u}_z$ (B) $\frac{6}{a}(x - y)$
(C) $\frac{6}{a} \mathbf{u}_z$ (D) $\frac{3}{a}(x - y)$

41. The point $P(2, 3, 1)$ lies on the planner boundary separating region 1 from region 2. The unit vector $\mathbf{u}_{N12} = 0.6\mathbf{u}_x + 0.48\mathbf{u}_y + 0.64\mathbf{u}_z$ is directed from region 1 to region 2. If $\mu_{r1} = 2$, $\mu_{r2} = 8$ and $\mathbf{H}_1 = 100\mathbf{u}_x - 300\mathbf{u}_y + 200\mathbf{u}_z \text{ A/m}$, then $\mathbf{H}_2$ is

- (A) $40.3\mathbf{u}_x + 48.3\mathbf{u}_y - 178.9\mathbf{u}_z \text{ A/m}$
(B) $80.2\mathbf{u}_x - 315.8\mathbf{u}_y + 178.9\mathbf{u}_z \text{ A/m}$
(C) $40.3\mathbf{u}_x - 315.8\mathbf{u}_y - 178.9\mathbf{u}_z \text{ A/m}$
(D) $80.2\mathbf{u}_x + 48.3\mathbf{u}_y + 178.9\mathbf{u}_z \text{ A/m}$

42. The plane separates air ($z > 0, \mu_r = 1$) from iron ($z \leq 0, \mu_r = 20$). In air magnetic field intensity is $\mathbf{H} = 10\mathbf{u}_x + 15\mathbf{u}_y - 3\mathbf{u}_z \text{ A/m}$. The magnetic flux density in iron is

- (A) $5.02\mathbf{u}_x + 7.5\mathbf{u}_y - 0.076\mathbf{u}_z \text{ mWb/m}^2$
(B) $12.6\mathbf{u}_x + 18.9\mathbf{u}_y - 75.4\mathbf{u}_z \mu\text{Wb/m}^2$
(C) $251\mathbf{u}_x + 377\mathbf{u}_y - 3.77\mathbf{u}_z \mu\text{Wb/m}^2$
(D) $251\mathbf{u}_x + 377\mathbf{u}_y - 1508\mathbf{u}_z \mu\text{Wb/m}^2$

43. The plane $2x + 3y - 4z = 1$ separates two regions. Let $\mu_{r1} = 2$ in region 1 defined by $2x + 3y - 4z > 1$, while $\mu_{r2} = 5$ in region 2 where $2x + 3y - 4z < 1$. In region $\mathbf{H}_1 = 50\mathbf{u}_x - 30\mathbf{u}_y + 20\mathbf{u}_z \text{ A/m}$. In region 2, $\mathbf{H}_2$ will be

- (A) $63.4\mathbf{u}_x + 43.18\mathbf{u}_y - 19.4\mathbf{u}_z \text{ A/m}$
(B) $52.9\mathbf{u}_x - 25.66\mathbf{u}_y + 14.2\mathbf{u}_z \text{ A/m}$
(C) $48.6\mathbf{u}_x - 16.4\mathbf{u}_y - 46.3\mathbf{u}_z \text{ A/m}$
(D) None of the above

SOLUTIONS

$$\begin{aligned}
 1. \text{ (C) } \mathbf{H} &= \int_{-\infty}^{\infty} \frac{Id\mathbf{L} \times \mathbf{u}_R}{4\pi R^2} \\
 &= \frac{I}{4\pi} \int_0^{\infty} \frac{-\mathbf{u}_y dy [2\mathbf{u}_x + (3-y)\mathbf{u}_y]}{[2^2 + (3-y)^2]^{3/2}} = \\
 &= \frac{I}{4\pi} \int_0^{\infty} \frac{2\mathbf{u}_z dy}{[2^2 + (3-y)^2]^{3/2}}
 \end{aligned}$$

$$3 - y = 2 \tan \theta, -dy = 2 \sec^2 \theta,$$

$$\theta_1 = 56.31^\circ, \theta_2 = -90^\circ$$

$$\begin{aligned}
 \mathbf{H} &= \frac{I}{4\pi} \int_{-90^\circ}^{56.31^\circ} \frac{2\mathbf{u}_z d\theta}{2 \sec^2 \theta} \\
 &= \frac{I}{4\pi} \mathbf{u}_z [\sin \theta]_{-90^\circ}^{56.31^\circ} = 145.8 \mathbf{u}_z \text{ mA/m}
 \end{aligned}$$

$$2. \text{ (A) } \mathbf{H} = \mathbf{H}_y + \mathbf{H}_z, \mathbf{H}_z = \frac{I_z}{2\pi\rho} \mathbf{u}_\phi$$

$$\rho = \sqrt{(-3)^2 + (4)^2} = 5$$

$$\mathbf{u}_\phi = \frac{-\mathbf{u}_z \times (-3\mathbf{u}_x + 4\mathbf{u}_y)}{5} = \frac{3\mathbf{u}_y + 4\mathbf{u}_x}{5}$$

$$\mathbf{H}_z = \frac{24}{2\pi(5)} \frac{(4\mathbf{u}_x + 3\mathbf{u}_y)}{5} = 0.611\mathbf{u}_x + 0.458\mathbf{u}_y \text{ mA/m}$$

$$\mathbf{H}_y = \frac{I_y}{2\pi\rho} \mathbf{u}_\phi, \rho = \sqrt{(-3)^2 + (5)^2} = \sqrt{34}$$

$$\mathbf{u}_\phi = \mathbf{u}_y \times \frac{(-3\mathbf{u}_x + 5\mathbf{u}_z)}{\sqrt{34}} = \frac{3\mathbf{u}_z - 5\mathbf{u}_x}{\sqrt{34}}$$

$$\mathbf{H}_y = \frac{12}{2\pi\sqrt{34}} \frac{(-5\mathbf{u}_x + 3\mathbf{u}_z)}{\sqrt{34}}$$

$$= -0.281\mathbf{u}_x + 0.168\mathbf{u}_z \text{ mA/m}$$

$$\mathbf{H} = \mathbf{H}_y + \mathbf{H}_z = 0.331\mathbf{u}_x + 0.458\mathbf{u}_y + 0.168\mathbf{u}_z \text{ mA/m}$$

$$3. \text{ (A) } \mathbf{H} = \frac{I}{2\pi\rho} \mathbf{u}_\phi - \int_{-4}^4 \frac{Idz\mathbf{u}_z \times (\rho\mathbf{u}_\rho - z\mathbf{u}_z)}{4\pi(\rho^2 + z^2)^{3/2}}$$

$$\frac{I}{2\pi\rho} \mathbf{u}_\phi - \frac{I}{4\pi} \int_{-4}^4 \frac{\rho dz}{4\pi(\rho^2 + z^2)^{3/2}} \mathbf{u}_\phi$$

$$= \frac{I}{2\pi\rho} \mathbf{u}_\phi - \frac{I}{4\pi} \frac{8}{\rho\sqrt{(\rho^2 + 16)}} \mathbf{u}_\phi$$

$$= \frac{I}{2\pi\rho} \left(1 - \frac{4}{\sqrt{(\rho^2 + 16)}} \right) \mathbf{u}_\phi$$

$$\text{At } \rho = -3, \phi = 60^\circ, I = 3\pi,$$

$$\mathbf{H} = 0.1\mathbf{u}_\phi \text{ A/m}$$

$$4. \text{ (B) } \mathbf{H} = \int \frac{Id\mathbf{L} \times \mathbf{u}_R}{4\pi R^2}, Id\mathbf{L} = 4dx\mathbf{u}_x$$

$$\mathbf{u}_R = \frac{-(1+x)\mathbf{u}_x + 3\mathbf{u}_y + 2\mathbf{u}_z}{R}$$

$$R = \sqrt{(1+x)^2 + 3^2 + 2^2} = \sqrt{x^2 + 2x + 14}$$

$$\mathbf{H} = \int_{-\infty}^{\infty} \frac{4dx\mathbf{u}_x \times [-(1+x)\mathbf{u}_x + 3\mathbf{u}_y + 2\mathbf{u}_z]}{4\pi(x^2 + 2x + 14)^{3/2}}$$

$$= \int_{-\infty}^{\infty} \frac{(12\mathbf{u}_z - 8\mathbf{u}_y)dx}{4\pi(x^2 + 2x + 14)^{3/2}} = \frac{2(12\mathbf{u}_z - 8\mathbf{u}_y)}{4\pi 13}$$

$$= 0.147\mathbf{u}_z - 0.098\mathbf{u}_y \text{ A/m}$$

$$5. \text{ (B) } \mathbf{H} = \frac{I}{2\pi\rho} \mathbf{u}_\phi - \int_{-a}^a \frac{Idz\mathbf{u}_z \times (\rho\mathbf{u}_\rho - z\mathbf{u}_z)}{4\pi(\rho^2 + z^2)^{3/2}}$$

$$= \frac{I}{2\pi\rho} \mathbf{u}_\phi - \int_{-a}^a \frac{\rho Idz\mathbf{u}_\phi}{4\pi(\rho^2 + z^2)^{3/2}}$$

$$\int_{-a}^a \frac{\rho Idz\mathbf{u}_\phi}{4\pi(\rho^2 + z^2)^{3/2}} = \frac{\rho I\mathbf{u}_\phi}{4\pi} \frac{z}{\rho^2(\rho^2 + z^2)^{3/2}} \Big|_{-a}^a = \frac{Ia\mathbf{u}_\phi}{2\pi\rho(\rho^2 + z^2)^{3/2}}$$

$$= \frac{I}{2\pi\rho} \left(1 - \frac{a}{\sqrt{(\rho^2 + a^2)}} \right) \mathbf{u}_\phi \text{ A/m}$$

$$\text{At } \rho = 1, H = \frac{I}{2\pi\rho}$$

$$\Rightarrow 1 - \frac{a}{\sqrt{1+a^2}} = \frac{1}{2}$$

$$\Rightarrow a = \frac{1}{\sqrt{3}} = 0.577 \text{ m}$$

$$6. \text{ (A) } \mathbf{H} = \int \frac{Id\mathbf{L} \times \mathbf{u}_R}{4\pi R^2}$$

$$= \int_0^{2\pi} \frac{Id\phi\mathbf{u}_\phi \times (-\mathbf{u}_\rho)}{4\pi a} = \frac{I}{2a} \mathbf{u}_z \text{ A/m}$$

$$I = 3 \text{ A}, a = 0.5 \text{ m}, \mathbf{H} = 3\mathbf{u}_z \text{ A/m}$$

$$7. \text{ (D) } \mathbf{H} = \iint \frac{\mathbf{K} \times \mathbf{u}_R dxdy}{4\pi R^2}$$

$$= \int_{-2}^2 \int_{-\infty}^{\infty} \frac{4\mathbf{u}_x \times (-x\mathbf{u}_x - y\mathbf{u}_y - 3\mathbf{u}_z) dx dy}{4\pi(x^2 + y^2 + 9)^{3/2}}$$

$$= \int_{-2}^2 \int_{-\infty}^{\infty} \frac{4(-y\mathbf{u}_z - 3\mathbf{u}_y) dx dy}{4\pi(x^2 + y^2 + 9)^{3/2}}$$

$$= \int_{-2}^2 \int_{-\infty}^{\infty} \frac{12\mathbf{u}_y dx dy}{4\pi(x^2 + y^2 + 9)^{3/2}}$$

$$= -\frac{3}{\pi} \mathbf{u}_y \int_{-2}^2 \frac{2}{y^2 + 9} dy$$

$$= -\frac{6}{\pi} \mathbf{u}_y \frac{1}{3} \tan^{-1} \left(\frac{y}{3} \right) \Big|_{-2}^2 = -0.75\mathbf{u}_y \text{ A/m}$$

$$\begin{aligned}
8. (D) \mathbf{H} &= \int \frac{Id\mathbf{L} \times \mathbf{u}_R}{4\pi R^2} \\
&= \int_0^\infty \frac{-Idz\mathbf{u}_z \times (-z\mathbf{u}_z + \mathbf{u}_y)}{4\pi(1+z^2)^{3/2}} + \int_0^\infty \frac{Idx\mathbf{u}_x \times (-x\mathbf{u}_x + \mathbf{u}_y)}{4\pi(1+x^2)^{3/2}} \\
&= \int_0^\infty \frac{Idz\mathbf{u}_x}{4\pi(1+z^2)^{3/2}} + \int_0^\infty \frac{Idx\mathbf{u}_z}{4\pi(1+x^2)^{3/2}} \\
&= \frac{I}{4\pi} \left(\left. \frac{z\mathbf{u}_x}{\sqrt{(1+z^2)}} \right|_0^\infty + \left. \frac{x\mathbf{u}_z}{\sqrt{(1+x^2)}} \right|_0^\infty \right) \\
&= \frac{I}{4\pi} (\mathbf{u}_x + \mathbf{u}_z) = 0.8(\mathbf{u}_x + \mathbf{u}_z) \text{ mA/m}
\end{aligned}$$

9. (A) Using Ampere's circuital law

$$\oint \mathbf{H} \cdot d\mathbf{L} = 2\pi\rho H_\phi = I_{encl}$$

$$\text{At } \rho = 0.5 \text{ cm, } I_{encl} = 10 \text{ mA}$$

$$2\pi(5 \times 10^{-3})H_\phi = 10, H_\phi = 0.32 \text{ A/m}$$

10. (B) At $\rho = 1.5 \text{ cm}$ enclosed current

$$I_{encl} = 10 + 2\pi(0.01)(400) = 35.13 \text{ mA}$$

$$2\pi(0.015)H_\phi = 35.13 \times 10^{-3} \Rightarrow H_\phi = 0.37 \text{ A/m}$$

11. (C) The enclosed current is

$$I_{encl} = 10 + 2\pi(0.01)400 - 2\pi(0.02)250 + 2\pi(0.03)300 = 60.3 \text{ mA/m}$$

$$2\pi(0.035)H_\phi = 60.3 \text{ M} \Rightarrow H_\phi = 0.27 \text{ A/m}$$

$$12. (A) \mathbf{H} = \frac{1}{2}(-30 - 40)\mathbf{u}_x \times (-\mathbf{u}_z) = -35\mathbf{u}_y \text{ A/m}$$

$$13. (B) \mathbf{H} = \frac{1}{2}\mathbf{K} \times \mathbf{u}_n$$

$$= \frac{1}{2}(30 - 40)\mathbf{u}_x \times (-\mathbf{u}_z) = -5\mathbf{u}_y \text{ A/m}$$

$$14. (A) \mathbf{H} = \frac{1}{2}(-30 + 40)\mathbf{u}_x + (-\mathbf{u}_z) = 5\mathbf{u}_y \text{ A/m}$$

$$15. (C) \mathbf{J} = \nabla \times \mathbf{H} = \frac{1}{\rho} \frac{\partial(\rho H_\phi)}{\partial \rho} \mathbf{u}_z$$

$$= \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(2 + \frac{\rho^2}{2} \right) \mathbf{u}_z = \mathbf{u}_z \text{ A/m}$$

$$16. (D) \mathbf{J} = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{3}{\rho} \right) = 0$$

$$17. (A) \mathbf{F} = e(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

$$\text{If } \mathbf{F} = 0, \mathbf{E} = -\mathbf{v} \times \mathbf{B} = \mathbf{B} \times \mathbf{v}$$

$$\begin{aligned}
&\begin{bmatrix} \mathbf{u}_x & \mathbf{u}_y & \mathbf{u}_z \\ 1 & 2 & 3 \\ 3 & 12 & -4 \end{bmatrix} \times 10^5 \times 10^{-3} \\
&= [\mathbf{u}_x(-8 - 36) - \mathbf{u}_y(-4 - 9) + \mathbf{u}_z(12 - 6)] \times 10^2 \text{ V/m} \\
&= -4.4\mathbf{u}_x + 1.3\mathbf{u}_y + 0.6\mathbf{u}_z \text{ kV/m}
\end{aligned}$$

$$18. (D) \mathbf{v}(0) \times \mathbf{B} = (2\mathbf{u}_x - 3\mathbf{u}_y - 4\mathbf{u}_z)10^5$$

$$\times (-3\mathbf{u}_x + 2\mathbf{u}_y - \mathbf{u}_z)10^{-3}$$

$$= 1100\mathbf{u}_x + 1400\mathbf{u}_y - 500\mathbf{u}_z$$

$$\mathbf{F}(0) = Q[\mathbf{E} + \mathbf{v} \times \mathbf{B}]$$

$$= 2 \times 10^{-16}[1200\mathbf{u}_x + 1200\mathbf{u}_y - 200\mathbf{u}_z]$$

$$= 4 \times 10^{-14}[6\mathbf{u}_x + 6\mathbf{u}_y - \mathbf{u}_z]$$

$$\mathbf{F} = m\mathbf{a} \Rightarrow \mathbf{a} = \frac{\mathbf{F}}{m} = \frac{4 \times 10^{-14}}{5 \times 10^{-26}}[6\mathbf{u}_x + 6\mathbf{u}_y - \mathbf{u}_z]$$

$$= 800[6\mathbf{u}_x + 6\mathbf{u}_y - \mathbf{u}_z]10^9 \text{ m/s}^2$$

$$19. (C) \mathbf{F} = e\mathbf{v} \times \mathbf{B}$$

$$= -(1.6 \times 10^{-19})(4.5 \times 10^7 \mathbf{u}_y)(2.5 \times 10^{-3} \mathbf{u}_z)$$

$$= -1.8 \times 10^{-14} \mathbf{u}_x \text{ N}$$

This force will be constant during the time the electron travels the field. It establishes a negative x -directed velocity as it leaves the field, given by the acceleration times the transit time t_t ,

$$v_x = \frac{Ft_t}{m} = \left(\frac{-1.8 \times 10^{-14}}{9.1 \times 10^{-31}} \right) \left(\frac{2.5 \times 10^{-2}}{4.5 \times 10^7} \right) = -1.1 \times 10^7 \text{ m/s}$$

$$t_{50} = \frac{0.5 - 0.025}{4.5 \times 10^7} = 1.06 \times 10^{-8} \text{ s}$$

In that time, the electron moves to an x coordinate given by

$$x = v_x t_{50} = -(1.1 \times 10^7)(1.06 \times 10^{-8}) = -0.117 \text{ m}$$

$$x = -11.7 \text{ cm, } z = 0$$

$$20. (A) \mathbf{F}_{BC} = \int_B^C I_{\text{loop}} d\mathbf{L} \times \mathbf{B}_{\text{from wire at BC}}$$

$$= \int_1^4 (6 \times 10^{-3}) dz \mathbf{u}_z \times \frac{15\mu_0}{2\pi(3)} \mathbf{u}_y = -1.8 \times 10^{-8} \mathbf{u}_x = -18\mathbf{u}_x \text{ nN}$$

21. (C) The field from the long wire now varies with position along the loop segment.

$$\begin{aligned}
\mathbf{F}_{AB} &= \int_1^3 (6 \times 10^{-3}) dx \mathbf{u}_x \times \frac{15\mu_0}{2\pi x} \mathbf{u}_y \\
&= \frac{45 \times 10^{-3}}{\pi} \mu_0 \ln 3 \mathbf{u}_z = 19.8\mathbf{u}_z \text{ nN}
\end{aligned}$$

22. (A) This will be the vector sum of the forces on the four sides. By symmetry, the forces on sides AB and CD will be equal and opposite, and so will cancel. This leaves the sum of forces on side BC and DA

$$\mathbf{F}_{DA} = \int_1^4 -(6 \times 10^{-3}) dx \mathbf{u}_z \times \frac{15\mu_0}{2\pi(1)} \mathbf{u}_y = 54 \mathbf{u}_x \text{ nN}$$

$$\mathbf{F}_{total} = \mathbf{F}_{DA} + \mathbf{F}_{BC} = (54 - 18) \mathbf{u}_x = 36 \mathbf{u}_x \text{ nN}$$

23. (A) $\mathbf{F} = \int Id\mathbf{L} \times \mathbf{B}$

$$= I \int_1^2 dx \mathbf{u}_x \times \mathbf{B} + I \int_1^2 dy \mathbf{u}_y \times \mathbf{B} + I \int_1^3 dx \mathbf{u}_x \times \mathbf{B} + I \int_2^1 dy \mathbf{u}_y \times \mathbf{B}$$

$$\mathbf{u}_x \times \mathbf{B} = \mathbf{u}_x [6x\mathbf{u}_x - 9y\mathbf{u}_y + 3z\mathbf{u}_z] = 3z\mathbf{u}_y - 9y\mathbf{u}_z$$

$$\mathbf{u}_y \times \mathbf{B} = \mathbf{u}_y [6x\mathbf{u}_x - 9y\mathbf{u}_y + 3z\mathbf{u}_z] = 3z\mathbf{u}_x - 6x\mathbf{u}_z$$

$z = 0$ for all element

$$\mathbf{F} = I \int_1^3 dx (-9y\mathbf{u}_z)_{y=1} + I \int_1^2 dy (-6x\mathbf{u}_z)_{x=3} + I \int_3^1 dx (-9y\mathbf{u}_z)_{y=2} + I \int_2^1 dy (-6x\mathbf{u}_z)_{x=1}$$

$$= I(-18 - 18 + 36 + 6)\mathbf{u}_z = 5 \times 6\mathbf{u}_z = 30\mathbf{u}_z \text{ N}$$

24. (B) Within the region $-1 < y < 1$, the magnetic fields from the two outer sheets (carrying $-4\mathbf{u}_z$ A/m) cancel, leaving only the field from the center sheet. Therefore $\mathbf{H} = -4\mathbf{u}_x$ A/m ($0 < y < 1$) and $\mathbf{H} = 4\mathbf{u}_x$ A/m ($-1 < y < 0$). Outside ($y > 1$ and $y < -1$) the fields from all three sheet cancel, leaving $H = 0$ ($y > 1$, $y < -1$). So at $x = 0$, $y = 0.5$

$$\frac{\mathbf{F}}{m} = I\mathbf{u}_z \times \mathbf{B} = (7 \times 10^{-3})\mathbf{u}_z \times -4\mu_0\mathbf{u}_x = -35.2\mathbf{u}_y \text{ nN/m}$$

25. (D) $\frac{\mathbf{F}}{m} = I\mathbf{u}_x \times (-4\mu_0\mathbf{u}_x) = 0$

26. (A) $\mathbf{F} = \int Id\mathbf{L} \times \mathbf{B}$

$$= \int_0^1 100 dz \mathbf{u}_z \times \frac{-100\mu_0\mathbf{u}_y}{2\pi(5 \times 10^{-3})} = 0.4\mathbf{u}_x \text{ N/m}$$

27. (B) The field from the current strip at the filament location

$$\mathbf{B} = \int_{0.5}^{1.5} \frac{6\mu_0\mathbf{u}_x}{2\pi y} dy = \frac{3\mu_0}{\pi} \ln\left(\frac{1.5}{0.5}\right) \mathbf{u}_x$$

$$= 1.32 \times 10^{-6} \mathbf{u}_x \text{ Wb/m}^2$$

$$\mathbf{F} = \int Id\mathbf{L} \times \mathbf{B}$$

$$= \int_0^1 5 dz \mathbf{u}_z \times 1.32 \times 10^{-6} \mathbf{u}_x dz = 6.6\mathbf{u}_y \text{ } \mu\text{N/m}$$

28. (A) $\mathbf{F} = \int_{area} \mathbf{K} \times \mathbf{B} dS = \int_0^1 \int_{0.5}^{1.5} 6\mathbf{u}_z \times \frac{-5\mu_0\mathbf{u}_x}{2\pi y} dy$

$$= -\frac{15\mu_0}{\pi} \ln\left(\frac{1.5}{0.5}\right) \mathbf{u}_y = -6.6\mathbf{u}_y \text{ } \mu\text{N/m}$$

29. (A) $\chi_m + 1 = \mu_r$, $\chi_m + 1 = 6.5$, $\chi_m = 5.5$

30. (A) $\mathbf{B} = \mu\mathbf{H} = \mu_0\mu_r\mathbf{H}$

$$= 4\pi \times 10^{-7} \times 6.5(10\mathbf{u}_x + 25\mathbf{u}_y - 40\mathbf{u}_z)$$

$$= 82\mathbf{u}_x + 204\mathbf{u}_y - 327\mathbf{u}_z \text{ } \mu\text{Wb/m}^2$$

31. (C) $\mathbf{M} = \chi_m\mathbf{H} = 55\mathbf{u}_x + 137.5\mathbf{u}_y - 220\mathbf{u}_z \text{ A/m}$

32. (B) $W = \frac{1}{2} \mathbf{H} \cdot \mathbf{B} = \frac{1}{2} \mu H^2$

$$= \frac{1}{2} \times 6.5 \times 4\pi \times 10^{-7} (100 + 625 + 1600) = 9.5 \text{ mJ/m}^2$$

33. (D) $\mu_r = \chi_m + 1 = 3.1 + 1 = 4.1$, $\mu = \mu_0\mu_r = 4.1\mu_0$

$$\mathbf{H} = \frac{\mathbf{B}}{\mu} = \frac{0.4y\mathbf{u}_z}{4.1 \times 4\pi \times 10^{-7}} = 77.6y\mathbf{u}_z \text{ kA/m}$$

34. (A) $\mathbf{M} = \chi_m\mathbf{H} = (3.1)(77.6)y\mathbf{u}_z = 241y\mathbf{u}_z \text{ kA/m}$

35. (C) For case 1, $\mu = \frac{B_1}{H} = \frac{2}{1200}$

$$\mu_{r1} = \frac{\mu}{\mu_0} = \frac{1}{600} \times \frac{1}{4\pi \times 10^{-7}} = 1326.3$$

$$\chi_m = \mu_r - 1 = 1325.3$$

$$M_1 = \chi_m H_1 = 1590 \times 10^6 \text{ A/m}$$

For case 2, $\mu = \frac{B_2}{H_2} = \frac{1.4}{400}$

$$\mu_{r1} = \frac{1.4}{400 \times 4\pi \times 10^{-7}} = 2785.2$$

$$\chi_m = 2784.2$$

$$M_2 = \chi_m H_2 = 1.114 \times 10^6 \text{ A/m}$$

$$\Delta M = (1590 - 1.114) \times 10^6 = 476 \text{ kA/m}$$

36. (B) $\mathbf{M} = N\mathbf{m} = (2.7 \times 10^{29})(2.6 \times 10^{-30} \mathbf{u}_y)$

$$= 0.7\mathbf{u}_y \text{ A/m}$$

$$\mathbf{H} = \frac{\mathbf{M}}{\mu_r - 1} = \frac{0.7\mathbf{u}_y}{4.2 - 1} = 0.22\mathbf{u}_y \text{ A/m}$$

37. (A) $\mathbf{M} = \frac{\mathbf{B}}{\mu_0} \left(\frac{1}{\chi_m} + 1 \right)^{-1}$

$$= \frac{0.02}{4\pi \times 10^{-7}} \left(\frac{1}{0.003} + 1 \right)^{-1} = 47.6 \text{ A/m}$$

$$\begin{aligned} 38. \text{ (A) } W &= \frac{1}{2} \mathbf{H} \cdot \mathbf{B} = \frac{1}{2} \mu_o H^2 \\ &= \frac{1}{2} (4\pi \times 10^{-7}) (600)^2 = 0.226 \text{ J/m}^3 \end{aligned}$$

$$\begin{aligned} 39. \text{ (A) } W &= \int_0^{H_o} H \cdot dB = \int_0^{H_o} H \left(\frac{1}{3} + 2H \right) dH \\ &= \frac{H_o^2}{6} + \frac{2H_o^3}{3} = 6.2 \text{ MJ/m}^3 \end{aligned}$$

$$40. \text{ (A) } \mathbf{J}_b = \nabla \times \mathbf{M} = \frac{12}{a} \mathbf{u}_z$$

$$41. \text{ (B) } \mathbf{B}_1 = 200\mu_o \mathbf{u}_x - 600\mu_o \mathbf{u}_y + 400\mu_o \mathbf{u}_z$$

Its normal component at the boundary is

$$\begin{aligned} \mathbf{B}_{1N} &= (\mathbf{B}_1 \cdot \mathbf{u}_{N12}) \mathbf{u}_{N12} \\ &= (52.8\mathbf{u}_x + 42.24\mathbf{u}_y + 56.32\mathbf{u}_z) \mu_o = \mathbf{B}_{2N} \\ \Rightarrow \mathbf{H}_{2N} &= \frac{\mathbf{B}_{2N}}{8\mu_o} = 6.60\mathbf{u}_x + 5.28\mathbf{u}_y + 7.04\mathbf{u}_z \\ \mathbf{H}_{1N} &= \frac{\mathbf{B}_{1N}}{12\mu_o} = 26.40\mathbf{u}_x + 21.12\mathbf{u}_y + 28.16\mathbf{u}_z \\ \mathbf{H}_{1T} &= \mathbf{H}_1 - \mathbf{H}_{1N} = (100\mathbf{u}_x - 300\mathbf{u}_y + 200\mathbf{u}_z) \\ &\quad - (26.4\mathbf{u}_x + 21.12\mathbf{u}_y + 28.16\mathbf{u}_z) \\ &= 73.6\mathbf{u}_x - 321.12\mathbf{u}_y + 171.84\mathbf{u}_z \\ \mathbf{H}_{1T} &= \mathbf{H}_{2T} \\ \mathbf{H}_2 &= \mathbf{H}_{2N} + \mathbf{H}_{2T} = 80.2\mathbf{u}_x - 315.8\mathbf{u}_y + 178.9\mathbf{u}_z \text{ A/m} \end{aligned}$$

$$42. \text{ (C) } \mathbf{H}_{N1} = -3\mathbf{u}_z, \mathbf{H}_{T1} = 10\mathbf{u}_x + 15\mathbf{u}_y$$

$$\mathbf{H}_{T2} = \mathbf{H}_{T1} = 10\mathbf{u}_x + 15\mathbf{u}_y$$

$$\mathbf{H}_{N2} = \frac{\mu_1}{\mu_2} \mathbf{H}_{N1} = \frac{1}{20} (-3\mathbf{u}_z) = 0.15\mathbf{u}_z$$

$$\mathbf{H}_2 = \mathbf{H}_{N2} + \mathbf{H}_{T2} = 10\mathbf{u}_x + 15\mathbf{u}_y - 0.15\mathbf{u}_z$$

$$\begin{aligned} \mathbf{B}_2 &= \mu_2 \mathbf{H}_2 = 20 \times 4\pi \times 10^{-7} (10\mathbf{u}_x + 15\mathbf{u}_y - 0.15\mathbf{u}_z) \\ &= 251\mathbf{u}_x + 377\mathbf{u}_y - 3.77\mathbf{u}_z \text{ } \mu\text{Wb/m}^2 \end{aligned}$$

43. (B) At the boundary normal unit vector

$$\mathbf{u}_n = \frac{\nabla(2x + 3y - 4z)}{|\nabla(2x + 3y - 4z)|} = \frac{2\mathbf{u}_x + 3\mathbf{u}_y - 4\mathbf{u}_z}{\sqrt{29}}$$

$$= 0.37\mathbf{u}_x + 0.56\mathbf{u}_y - 0.74\mathbf{u}_z$$

Since this vector is found through the gradient, it will point in the direction of increasing values of $2x + 3y - 4z$, and so will be directed into region 1. Thus

$$\mathbf{u}_n = \mathbf{u}_{n21} \text{ .}$$

The normal component of $\mathbf{H}_1$ is

$$\mathbf{H}_{N1} = (\mathbf{H}_1 \cdot \mathbf{u}_{N21}) \mathbf{u}_{N21}$$

$$\begin{aligned} \mathbf{H}_1 \cdot \mathbf{u}_{N21} &= (50\mathbf{u}_x - 30\mathbf{u}_y + 20\mathbf{u}_z) \cdot (0.37\mathbf{u}_x + 0.56\mathbf{u}_y - 0.74\mathbf{u}_z) \\ &= 18.5 - 16.8 - 14.8 = -13.1 \end{aligned}$$

$$\begin{aligned} (\mathbf{H}_1 \cdot \mathbf{u}_{N21}) \mathbf{u}_{N21} &= (-13.1)(0.37\mathbf{u}_x + 0.56\mathbf{u}_y - 0.74\mathbf{u}_z) \\ &= -4.83\mathbf{u}_x - 7.24\mathbf{u}_y + 9.66\mathbf{u}_z \text{ A/m} \end{aligned}$$

Tangential component of $\mathbf{H}_1$ at the boundary

$$\mathbf{H}_{T1} = \mathbf{H}_1 - \mathbf{H}_{N1}$$

$$\begin{aligned} &= (50\mathbf{u}_x - 30\mathbf{u}_y + 20\mathbf{u}_z) - (-4.83\mathbf{u}_x - 7.24\mathbf{u}_y + 9.66\mathbf{u}_z) \\ &= 54.83\mathbf{u}_x - 22.76\mathbf{u}_y + 10.34\mathbf{u}_z \text{ A/m} \end{aligned}$$

$$\mathbf{H}_{T2} = \mathbf{H}_{T1}$$

$$\mathbf{H}_{N2} = \frac{\mu_{r1}}{\mu_{r2}} \mathbf{H}_{N1} = \frac{2}{5} (-4.83\mathbf{u}_x - 7.24\mathbf{u}_y + 9.66\mathbf{u}_z)$$

$$= -1.93\mathbf{u}_x - 2.90\mathbf{u}_y + 3.86\mathbf{u}_z \text{ A/m}$$

$$\begin{aligned} \mathbf{H}_2 &= \mathbf{H}_{T2} + \mathbf{H}_{N2} = (54.83\mathbf{u}_x - 22.76\mathbf{u}_y + 10.34\mathbf{u}_z) \\ &\quad + (-1.93\mathbf{u}_x - 2.9\mathbf{u}_y + 3.86\mathbf{u}_z) \end{aligned}$$

$$= 52.9\mathbf{u}_x - 25.66\mathbf{u}_y + 14.2\mathbf{u}_z$$

Statement for Q.8–9:

The location of the sliding bar in fig. P8.4.8–9 is given by $x = 5t + 4t^3$. The separation of the two rails is 30 cm. Let $\mathbf{B} = x^2 \mathbf{u}_z$ T.

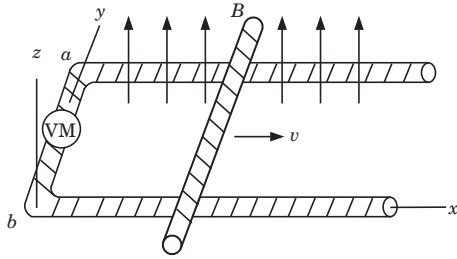


Fig. P8.4.8–9.

8. The voltmeter reading at $t = 0.5$ s is
 (A) -21.6 V (B) 21.6 V
 (C) -6.3 V (D) 6.3 V
9. The voltmeter reading at $x = 0.6$ m is
 (A) -1.68 V (B) 1.68 V
 (C) -0.933 V (D) 0.933 V

Statement for Q.10–11:

A perfectly conducting filament containing a 250Ω resistor is formed into a square as shown in fig. P8.4.10–11.

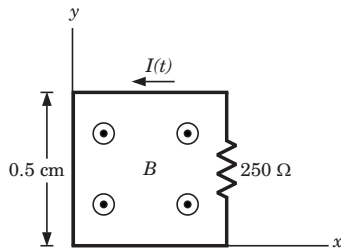


Fig. P8.4.10–11

10. If $\mathbf{B} = 6 \cos(120\pi t - 30^\circ) \mathbf{u}_z$ T, then the value of $I(t)$ is
 (A) $2.26 \sin(120\pi t - 30^\circ)$ A
 (B) $2.26 \cos(120\pi t - 30^\circ)$ A
 (C) $-2.26 \sin(120\pi t - 30^\circ)$ A
 (D) $-2.26 \cos(120\pi t - 30^\circ)$ A
11. If $\mathbf{B} = 2 \cos \pi(ct - y) \mathbf{u}_z$ μ T, where c is the velocity of light, then $I(t)$ is
 (A) $1.2(\cos \pi ct - \sin \pi ct)$ μ A
 (B) $1.2(\cos \pi ct - \sin \pi ct)$ mA
 (C) $1.2(\sin \pi ct - \sin \pi ct)$ μ A
 (D) $1.2(\sin \pi ct - \sin \pi ct)$ mA

Statement for Q.12–13:

Consider the fig. P8.4.12–13. The rails have a resistance of $2 \Omega/\text{m}$. The bar moves to the right at a constant speed of 9 m/s in a uniform magnetic field of 0.8 T. The bar is at $x = 2$ m at $t = 0$.

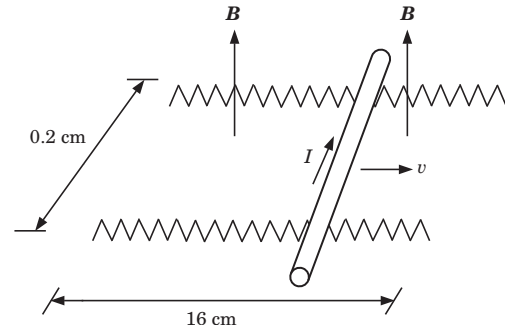


Fig. P8.4–12–14

12. If 6Ω resistor is present across the left-end with the right end open-circuited, then at $t = 0.5$ sec the current I is
 (A) -45 mA (B) 45 mA
 (C) -60 mA (D) 60 mA
13. If 6Ω resistor is present across each end, then I at 0.5 sec is
 (A) -12.3 mA (B) 12.3 mA
 (C) -7.77 mA (D) 77.7 mA

Statement for Q.14–15:

The internal dimension of a coaxial capacitor is $a = 1.2$ cm, $b = 4$ cm and $c = 40$ cm. The homogeneous material inside the capacitor has the parameter $\epsilon = 10^{-11}$ F/m, $\mu = 10^{-5}$ H/m and $\sigma = 10^{-5}$ S/m. The electric field intensity is $\mathbf{E} = \frac{10^7}{\rho} \cos(10^5 t) \mathbf{u}_\rho$ V/m.

14. The current density $\mathbf{J}$ is
 (A) $\frac{200}{\rho} \sin(10^5 t) \mathbf{u}_\rho$ A/m²
 (B) $\frac{400}{\rho} \sin(10^5 t) \mathbf{u}_\rho$ A/m²
 (C) $\frac{100}{\rho} \cos(10^5 t) \mathbf{u}_\rho$ A/m²
 (D) None of the above

15. The quality factor of the capacitor is
 (A) 0.1 (B) 10
 (C) 0.2 (D) 20

16. The following fields exist in charge free regions

$$\mathbf{P} = 60 \sin(\omega t + 10x) \mathbf{u}_z$$

$$\mathbf{Q} = \frac{10}{\rho} \cos(\omega t - 2\rho) \mathbf{u}_\phi$$

$$\mathbf{R} = 3\rho^2 \cot \phi \mathbf{u}_\rho + \frac{1}{\rho} \cos \phi \mathbf{u}_\phi$$

$$\mathbf{S} = \frac{1}{r} \sin \theta \sin(\omega t - 6r) \mathbf{u}_\theta$$

The possible electromagnetic fields are

- (A) $\mathbf{P}, \mathbf{Q}$ (B) $\mathbf{R}, \mathbf{S}$
(C) $\mathbf{P}, \mathbf{R}$ (D) $\mathbf{Q}, \mathbf{S}$

17. A parallel-plate capacitor with plate area of 5 cm^2 and plate separation of 3 mm has a voltage $50 \sin(10^3 t) \text{ V}$ applied to its plates. If $\epsilon_r = 2$, the displacement current is

- (A) $148 \cos(10^{10} t) \text{ nA}$ (B) $261 \cos(10^{10} t) \mu\text{A}$
(C) $261 \cos(10^{10} t) \text{ nA}$ (D) $148 \cos(10^{10} t) \mu\text{A}$

18. In a coaxial transmission line ($\epsilon_r = 1$), the electric field intensity is given by

$$\mathbf{E} = \frac{100}{\rho} \cos(10^9 t - 6z) \mathbf{u}_\rho \text{ V/m.}$$

The displacement current density is

- (A) $-\frac{100}{\rho} \sin(10^9 t - 6z) \mathbf{u}_\rho \text{ A/m}^2$
(B) $\frac{116}{\rho} \sin(10^9 t - 6z) \mathbf{u}_\rho \text{ A/m}^2$
(C) $-\frac{0.9}{\rho} \sin(10^9 t - 6z) \mathbf{u}_\rho \text{ A/m}^2$
(D) $-\frac{216}{\rho} \cos(10^9 t - 6z) \mathbf{u}_\rho \text{ A/m}^2$

Statement for Q.19–21:

Consider the region defined by $|x|, |y|$ and $|z| < 1$. Let $\epsilon = 5\epsilon_0$, $\mu = 4\mu_0$, and $\sigma = 0$ the displacement current density $\mathbf{J}_d = 20 \cos(1.5 \times 10^8 t - ax) \mathbf{u}_y \mu\text{A/m}^2$. Assume no DC fields are present.

- 19.** The electric field intensity $\mathbf{E}$ is
(A) $6 \sin(1.5 \times 10^8 t - ax) \mathbf{u}_y \text{ mV/m}$
(B) $6 \cos(1.5 \times 10^8 t - ax) \mathbf{u}_y \text{ mV/m}$
(C) $3 \cos(1.5 \times 10^8 t - ax) \mathbf{u}_y \text{ mV/m}$
(D) $3 \sin(1.5 \times 10^8 t - ax) \mathbf{u}_y \text{ mV/m}$

- 20.** The magnetic field intensity is
(A) $-4a \sin(1.5 \times 10^8 t - ax) \mathbf{u}_z \mu\text{A/m}$

- (B) $-4a \sin(1.5 \times 10^8 t - ax) \mathbf{u}_z \text{ mA/m}$
(C) $4a \sin(1.5 \times 10^8 t - ax) \mathbf{u}_z \mu\text{A/m}$
(D) $4a \sin(1.5 \times 10^8 t - ax) \mathbf{u}_z \text{ mA/m}$

21. The value of a is

- (A) 4.3 (B) 2.25
(C) 5 (D) 6

Statement for Q.22–23:

Let $\mathbf{H} = 2 \cos(10^{10} t - \beta x) \mathbf{u}_z \text{ A/m}$, $\mu = 3 \times 10^{-5} \text{ H/m}$, $\epsilon = 1.2 \times 10^{-10} \text{ F/m}$ and $\sigma = 0$ everywhere.

22. The electric flux density $\mathbf{D}$ is

- (A) $120 \cos(10^{10} t - \beta x) \text{ nC/m}^2$
(B) $-120 \cos(10^{10} t - \beta x) \text{ nC/m}^2$
(C) $120 \cos(10^{10} t + \beta x) \text{ nC/m}^2$
(D) None of the above

23. The magnetic flux density $\mathbf{B}$ is

- (A) $6.67 \times 10^4 \cos(10^{10} t + \beta x) \text{ T}$
(B) $6.67 \times 10^4 \cos(10^{10} t - \beta x) \text{ T}$
(C) $6 \times 10^{-5} \cos(10^{10} t + \beta x) \text{ T}$
(D) $6 \times 10^{-5} \cos(10^{10} t - \beta x) \text{ T}$

Statement for Q.24–25:

A material has $\sigma = 0$ and $\epsilon_r = 1$. The magnetic field intensity is $\mathbf{H} = 4 \cos(10^6 t - 0.01z) \mathbf{u}_y \text{ A/m}$.

24. The electric field intensity $\mathbf{E}$ is

- (A) $4.52 \sin(10^6 t - 0.01z) \text{ kV/m}$
(B) $4.52 \sin(10^6 t - 0.01z) \text{ V/m}$
(C) $4.52 \cos(10^6 t - 0.01z) \text{ V/m}$
(D) $4.52 \cos(10^6 t - 0.01z) \text{ kV/m}$

25. The value of μ_r is

- (A) 2 (B) 3
(C) 4 (D) 16

26. The surface $\rho = 3$ and 10 mm , and $z = 0$ and 25 cm are perfect conductors. The region enclosed by these surface has $\mu = 2.5 \times 10^{-6} \text{ H/m}$, $\epsilon = 4 \times 10^{-11} \text{ F/m}$ and $\sigma = 0$. If $\mathbf{H} = \frac{2}{\rho} \cos 8\pi z \cos \omega t \mathbf{u}_\phi \text{ A/m}$, then the value of ω is

- (A) $2\pi \times 10^6 \text{ rad/s}$ (B) $8\pi \times 10^6 \text{ rad/s}$
(C) $2\pi \times 10^8 \text{ rad/s}$ (D) $8\pi \times 10^8 \text{ rad/s}$

27. For distilled water $\mu = \mu_o$, $\varepsilon = 81\varepsilon_o$, and $\sigma = 2 \times 10^{-3}$ S/m, the ratio of conduction current density to displacement current density at 1 GHz is

- (A) 1.11×10^{-5} (B) 4.44×10^{-4}
(C) 2.68×10^{-6} (D) 1.68×10^{-7}

28. A conductor with cross-sectional area of 10 cm^2 carries a conductor current $2 \sin(10^9 t) \text{ mA}$. If $\sigma = 2.5 \times 10^6 \text{ S/m}$ and $\varepsilon_r = 4.6$, the magnitude of the displacement current density is

- (A) $48.4 \mu\text{A/m}^2$ (B) 8.11 nA/m^2
(C) 32.6 nA/m^2 (D) $16.4 \mu\text{A/m}^2$

29. In a certain region

$$\mathbf{J} = (4y\mathbf{u}_x + 2xz\mathbf{u}_y + z^3\mathbf{u}_z) \sin(10^4 t) \text{ A/m}$$

If volume charge density ρ_v in $z=0$ plane is zero, then ρ_v is

- (A) $3z^2 \cos(10^4 t) \text{ mC/m}^3$
(B) $0.3z^2 \cos(10^4 t) \text{ mC/m}^3$
(C) $-3z^2 \cos(10^4 t) \text{ mC/m}^3$
(D) $-0.3z^2 \cos(10^4 t) \text{ mC/m}^3$

30. In a charge-free region ($\sigma=0$, $\varepsilon = \varepsilon_o \varepsilon_r$, $\mu = \mu_o$) magnetic field intensity is $\mathbf{H} = 10 \cos(10^{11} t - 4y) \mathbf{u}_z \text{ A/m}$. The displacement current density is

- (A) $-40 \sin(10^9 t - 4y) \mathbf{u}_y \text{ A/m}$
(B) $40 \sin(10^9 t - 4y) \mathbf{u}_y \text{ A/m}$
(C) $-40 \sin(10^3 t - 4y) \mathbf{u}_x \text{ A/m}$
(D) $40 \sin(10^9 t - 4y) \mathbf{u}_x \text{ A/m}$

31. In a nonmagnetic medium ($\varepsilon_r = 6.25$) the magnetic field of an EM wave is $\mathbf{H} = 6 \cos \beta x \cos(10^8 t) \mathbf{u}_z \text{ A/m}$. The corresponding electric field is

- (A) $903 \sin(0.83x) \sin(10^8 t) \text{ V/m}$
(B) $903 \sin(12x) \sin(10^8 t) \text{ V/m}$
(C) $903 \sin(0.83x) \cos(10^8 t) \text{ V/m}$
(D) $903 \sin(12x) \cos(10^8 t) \text{ V/m}$

32. In a nonmagnetic medium

$$\mathbf{E} = 5 \cos(10^9 t - 8x) \mathbf{u}_x + 4 \sin(10^9 t - 8x) \mathbf{u}_z \text{ V/m}$$

The dielectric constant of the medium is

- (A) 3.39 (B) 1.84
(C) 5.76 (D) 2.4

SOLUTIONS

$$1. \text{ (B) } \text{emf} = -\frac{d\Phi}{dt} = -\frac{d}{dt} \int \mathbf{B} \cdot d\mathbf{S}$$

$$= 2\pi(0.2)^2(20)(377) \sin 377t \text{ mV} = 0.95 \cos 377t \text{ V}$$

$$2. \text{ (A) } \text{emf} = \frac{1}{2} B_o \omega L^2 = \frac{1}{2} (4)(2)(2)^2 = 16 \text{ V}$$

3. (C) Since B is constant over the loop area, the flux is $\Phi = \pi(0.1)^2 B = 0.31 \cos(120\pi t)$

$$\text{emf} = V_{ba}(t) = -\frac{d\Phi}{dt}$$

$$= 0.31(120\pi) \sin(120\pi t) = 118.43 \sin(120\pi t)$$

$$V_{ab} = -118.43 \sin(120\pi t)$$

$$4. \text{ (D) } I = \frac{V_{ab}}{R} = \frac{118.43 \sin(120\pi t)}{250} = 0.47 \sin(120\pi t)$$

$$5. \text{ (A) } \text{emf} = -\frac{d\Phi}{dt} = -\frac{d}{dt} \iint_{\text{loop area}} \mathbf{B} \cdot \mathbf{u}_z dz$$

$$= \frac{d}{dt} (3)(4)(6) \cos 5000t = -360000 \sin 5000t$$

$$I = \frac{\text{emf}}{R} = -\frac{360000 \sin 5000t}{900 \times 10^3} = -0.4 \sin 5000t \text{ A}$$

$$6. \text{ (C) } \Phi = \int_0^1 \int_0^1 20\mu_o \cos(3 \times 10^8 t - y) dx dy$$

$$= [20\mu_o \sin(3 \times 10^8 t - y)]_0^1$$

$$= 20\mu_o [\sin(3 \times 10^8 t - 1) - \sin(3 \times 10^8 t)] \text{ Wb}$$

$$\text{Emf} = -\frac{d\Phi}{dt}$$

$$= -20 \times (4\pi \times 10^{-7})(3 \times 10^8) \times [\cos(3 \times 10^8 t - 1) - \cos(3 \times 10^8 t)]$$

$$= 7540 [\cos(3 \times 10^8 t) - \cos(3 \times 10^8 t - 1)] \text{ V}$$

$$7. \text{ (D) In this case } \Phi = [20\mu_o (2\pi) \sin(3 \times 10^8 t - y)]_0^{2\pi} = 0$$

$$8. \text{ (A) } \Phi = \iint_{\text{area}} \mathbf{B} \cdot d\mathbf{S} = \int_0^{0.3} \int_0^x \tau^2 d\tau dy$$

$$= 0.1x^3 = 0.1(5t + 4t^3)^3 \text{ Wb}$$

$$\text{emf} = -\frac{d\Phi}{dt} = -0.1(3)(5t + 4t^3)^2(5 + 12t^2)$$

$$\text{At } t=0.5 \text{ s, } \text{emf} = -0.1(3)(2.5 + 0.5)^2(5 + 3) = -21.6 \text{ V}$$

$$9. \text{ (C) At } x=0.6 \text{ m, } 0.6 = 5t + 4t^3 \Rightarrow t = 0.119 \text{ s}$$

$$\text{At } t=0.119 \text{ s, } \text{emf} = -0.933 \text{ V}$$

$$10. (A) \Phi = \iint_{\text{area}} \mathbf{B} \cdot d\mathbf{S} = 6(0.5)^2 \cos(120\pi t - 30^\circ) \text{ Wb}$$

$$\text{emf} = -\frac{d\Phi}{dt} = 6(0.5)^2(120\pi) \sin(120\pi t - 30^\circ)$$

$$\begin{aligned} \text{The current is } \frac{\text{emf}}{R} &= \frac{6(0.5)^2(120\pi)}{250} \sin(120\pi t - 30^\circ) \text{ A} \\ &= 2.26 \sin(120\pi t - 30^\circ) \text{ A} \end{aligned}$$

$$\begin{aligned} 11. (D) \Phi &= \iint_{\text{area}} \mathbf{B} \cdot d\mathbf{S} = (0.5)(2) \int_0^{0.5} \cos(\pi y - \pi ct) dy \\ &= \frac{1}{\pi} \left[\sin\left(\pi ct - \frac{\pi}{2}\right) - \sin \pi ct \right] = \frac{1}{\pi} [-\cos \pi ct - \sin \pi ct] \text{ } \mu\text{Wb} \end{aligned}$$

$$\text{emf} = -\frac{d\Phi}{dt} = c [\cos \pi ct - \sin \pi ct] \text{ } \mu\text{V}$$

$$\begin{aligned} I(t) &= \frac{\text{emf}}{R} = \frac{3 \times 10^8}{250} [\cos \pi ct - \sin \pi ct] \text{ } \mu\text{A} \\ &= 12 [\cos \pi ct - \sin \pi ct] \text{ A} \end{aligned}$$

12. (A) The flux in the left-hand closed loop is

$$\Phi_l = B \times \text{area} = (0.8)(0.2)(2 + 9t)$$

$$\text{emf}_l = -\frac{d\Phi_l}{dt} = -(0.16)(9) = -1.44 \text{ V}$$

While the bar is in motion, the loop resistance is increasing with time,

$$R_l = 6 + 2[2(2 + 9t)]\Omega, \text{ At } t = 0.5, \quad R_l = 32 \Omega$$

$$I_l = -\frac{1.44}{32} = -45 \text{ mA}$$

13. (C) In this case, there will be contribution to the current from the right loop, which is now closed. The flux in the right loop, whose area decreases with time, is $\Phi_r = (0.8)(0.2)(16 - 2 - 9t)$

$$\text{emf}_r = -\frac{d\Phi_r}{dt} = 1.44 \text{ V}$$

$$R_r = 6 + 2(2(14 - 9t)), \text{ At } 0.5 \text{ s}, \quad R_r = 44 \Omega$$

The contribution to the current from the right loop

$$I_r = \frac{-1.44}{44} = 0.327 \text{ mA}$$

$$\text{The total current} = -32.7 - 45 = -77.7 \text{ mA}$$

$$14. (C) \mathbf{J} = \sigma \mathbf{E} = \frac{100}{\rho} \cos(10^5 t) \mathbf{u}_\rho \text{ A/m}^2$$

15. (A) Total conduction current

$$I_c = \iint \mathbf{J} \cdot d\mathbf{S} = 2\pi \rho l J = 2\pi \rho l \frac{100}{\rho} \cos(10^5 t) \mathbf{u}_\rho \text{ A/m}^2$$

$$= 80\pi \cos(10^5 t) \text{ A}$$

$$\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t} = \frac{\partial \epsilon \mathbf{E}}{\partial t} = -\frac{10}{\rho} \sin(10^5 t) \text{ A/m}^2$$

$$I_d = 2\pi \rho l \mathbf{J}_d = -2\pi l (10) \sin(10^5 t) = -8\pi \sin(10^5 t) \text{ A}$$

$$\text{Quality factor } \frac{|I_d|}{|I_c|} = \frac{8}{80} = 0.1$$

$$16. (A) \nabla \cdot \mathbf{P} = 0, \quad \nabla \times \mathbf{P} = -\frac{\partial P_z}{\partial x} \mathbf{u}_y \neq 0$$

$\mathbf{P}$ is a possible EM field

$$\nabla \cdot \mathbf{Q} = 0, \quad \nabla \times \mathbf{Q} = \frac{1}{\rho} \frac{\partial}{\partial \rho} [10 \cos(\omega t - 2\rho)] \mathbf{u}_z \neq 0$$

$\mathbf{Q}$ is a possible EM field

$$\nabla \cdot \mathbf{R} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (3\rho^2 \cot \phi) \cdot \frac{\sin \phi}{\rho} \neq 0, \mathbf{R} \text{ is not an EM field.}$$

$$\nabla \cdot \mathbf{S} = \frac{1}{r^2 \sin \theta} \sin(\omega t - 6r) \frac{\partial(\sin^2 \phi)}{\partial r} \neq 0$$

$\mathbf{S}$ is not an EM field. Hence (A) is correct.

$$17. (A) D = \epsilon E = \epsilon \frac{V}{d} \Rightarrow J_d = \frac{dD}{dt} = \frac{\epsilon}{d} \frac{dV}{dt}$$

$$\begin{aligned} I_d &= J \cdot S = \frac{\epsilon S}{d} \frac{dV}{dt} = \frac{2\epsilon_0 5 \times 10^{-4}}{3 \times 10^{-3}} 10^3 \times 50 \cos(10^3 t) \\ &= 148 \cos(10^{10} t) \text{ nA} \end{aligned}$$

18. (C)

$$\begin{aligned} \mathbf{J}_d &= \frac{\partial \mathbf{D}}{\partial t} = \epsilon \frac{\partial \mathbf{E}}{\partial t} = \epsilon_0 \frac{100}{\rho} [-\sin(10^9 t - 6z)] 10^9 \mathbf{u}_\rho \text{ A/m}^2 \\ &= -\frac{0.9}{\rho} \sin(10^9 t - 6z) \mathbf{u}_\rho \text{ A/m}^2 \end{aligned}$$

$$19. (D) \mathbf{D} = \int \mathbf{J}_d dt + C_1 = \frac{20 \times 10^{-6}}{15 \times 10^8} \sin(15 \times 10^8 - ax) \mathbf{u}_y$$

$$= 1.33 \times 10^{-13} \sin(15 \times 10^8 t - ax) \mathbf{u}_y \text{ C/m}^3$$

C_1 is set to zero since no DC fields are present.

$$\mathbf{E} = \frac{\mathbf{D}}{\epsilon} = \frac{1.33 \times 10^{-13}}{5\epsilon_0} \sin(15 \times 10^8 - ax) \mathbf{u}_y$$

$$= 3 \times 10^{-3} \sin(15 \times 10^8 t - ax) \text{ V/m}$$

$$20. (D) \nabla \times \mathbf{E} = \frac{\partial E_y}{\partial x} \mathbf{u}_z = -\frac{\partial \mathbf{B}}{\partial t}$$

$$= -a(3 \times 10^{-3}) \cos(15 \times 10^8 t - ax) \mathbf{u}_z = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\mathbf{B} = \frac{a(3 \times 10^{-3})}{15 \times 10^8} \sin(15 \times 10^8 t - ax) \mathbf{u}_z$$

$$\mathbf{H} = \frac{\mathbf{B}}{\mu} = \frac{2 \times 10^{-11}}{4 \times 4\pi \times 10^{-7}} \sin(15 \times 10^8 t - ax) \mathbf{u}_z \text{ A/m}$$

$$= 4 \times 10^{-6} a \sin(15 \times 10^8 t - ax) \mathbf{u}_z \text{ mA/m}$$

$$21. (B) \nabla \times \mathbf{H} = -\frac{\partial H_z}{\partial x} \mathbf{u}_y = \mathbf{J}_d$$

$$= a^2 (4 \times 10^{-6}) \cos (1.5 \times 10^8 t - ax) = J_D$$

Comparing the result

$$a^2 4 \times 10^{-6} = 20 \times 10^{-6}, \quad a = \sqrt{5} = 2.25$$

$$22. \text{ (B) } \nabla \times \mathbf{H} = -\frac{\partial H_z}{\partial x} \mathbf{u}_y = \frac{\partial \mathbf{D}}{\partial t}$$

$$\frac{\partial \mathbf{D}}{\partial t} = 2\beta \sin (10^{10} t - \beta x) \mathbf{u}_y$$

$$\mathbf{D} = -\frac{2\beta}{10^{10}} \cos (10^{10} t - \beta x) \mathbf{u}_y \text{ C/m}^2$$

$$\beta = \frac{\omega}{v} = 10^{10} \sqrt{\mu\epsilon} = 10^{10} \sqrt{3 \times 10^{-5} \times 12 \times 10^{-10}} = 600$$

$$\mathbf{D} = -120 \cos (10^{10} t - \beta x) \mathbf{u}_y \text{ nC/m}^2$$

$$23. \text{ (D) } \mathbf{B} = \mu \mathbf{H} = 6 \times 10^{-5} \cos (10^{10} t - \beta x) \mathbf{u}_z \text{ T}$$

$$24. \text{ (A) } \nabla \times \mathbf{H} = -\frac{\partial H_y}{\partial z} \mathbf{u}_x$$

$$\Rightarrow \nabla \times \mathbf{H} = 0.04 \cos (10^6 t - 0.01z) \mathbf{u}_x = \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

$$\mathbf{E} = \frac{0.04 \sin (10^6 t - 0.01z) \mathbf{u}_x}{10^6 \epsilon_0}$$

$$= 452 \sin (10^6 t - 0.01z) \mathbf{u}_x \text{ kV/m}$$

$$25. \text{ (B) } \nabla \times \mathbf{E} = \frac{\partial E_x}{\partial z} \mathbf{u}_y = -\mu \frac{\partial \mathbf{H}}{\partial t}$$

$$\frac{-0.04(0.01)}{10^6 \epsilon_0} \cos (10^6 t - 0.01z) \mathbf{u}_y = -\mu_r \mu_0 \frac{\partial \mathbf{H}}{\partial t}$$

$$\mathbf{H} = \frac{0.04(0.01)}{(10^6)(10^6) \mu_r \mu_0 \epsilon_0} \sin (10^6 t - 0.01z) \mathbf{u}_y$$

$$\frac{0.04(0.01)}{10^{12} \mu_r \mu_0 \epsilon_0} = 4 \Rightarrow \mu_r = \frac{(0.04)(0.01)}{4(10^{12})} \times (3 \times 10^8)^2 = 9$$

$$26. \text{ (D) } \nabla \times \mathbf{H} = -\frac{\partial H_\phi}{\partial z} \mathbf{u}_\rho$$

$$= \frac{16\pi}{\rho} \sin (8\pi z) \cos (\omega t) \mathbf{u}_\rho = \epsilon \frac{\partial \mathbf{E}}{\partial t}$$

$$\mathbf{E} = \frac{16\pi}{\rho \epsilon \omega} \sin (8\pi z) \sin (\omega t) \mathbf{u}_\rho$$

$$\nabla \times \mathbf{E} = \frac{\partial E_\rho}{\partial z} \mathbf{u}_\phi = \frac{(16\pi)(8\pi)}{\rho \epsilon \omega} \cos (8\pi z) \sin (\omega t) \mathbf{u}_\phi = -\mu \frac{\partial \mathbf{H}}{\partial t}$$

$$\mathbf{H} = \frac{128\pi^2}{\rho \epsilon \omega^2} \cos (8\pi z) \cos (\omega t) \mathbf{u}_\phi$$

This result must be equal to the given $\mathbf{H}$ field. Thus

$$\frac{128\pi^2}{\rho \epsilon \omega^2} = \frac{2}{\rho} \Rightarrow \omega = \frac{8\pi}{\sqrt{\mu\epsilon}} = \frac{8\pi}{\sqrt{4 \times 10^{-11} \times 2.5 \times 10^{-6}}}$$

$$= 8\pi \times 10^8 \text{ rad/s}$$

$$27. \text{ (B) At high frequency } \frac{J_c}{J_d} = \frac{\sigma E}{\omega \epsilon E} = \frac{\sigma}{\omega \epsilon}$$

$$= \frac{2 \times 10^{-3}}{2\pi \times 10^9 \times 81 \times \epsilon_0} = 4.44 \times 10^{-4}$$

$$28. \text{ (C) } J_c = \frac{I_c}{S} = \sigma E \Rightarrow E = \frac{I_c}{\sigma S}$$

$$\Rightarrow J_d = \epsilon \frac{\partial E}{\partial t} = \frac{\epsilon}{\sigma S} \frac{\partial I_c}{\partial t}$$

$$\Rightarrow J_d = \frac{4.6\epsilon_0(10^9)}{2.5 \times 10^6 \times 10 \times 10^{-4}} 2 \cos (10^9 t) 10^{-3}$$

$$|J_d| = 32.6 \text{ nA/m}^2$$

$$29. \text{ (B) } \nabla \cdot \mathbf{J} = (0 + 0 + 3z^2) \sin (10^4 t) = -\frac{8\rho}{\partial t}$$

$$\rho_v = \frac{3z^2 \cos (10^4 t)}{10^4} + C_1$$

$$\text{At } z=0, \quad \rho_v=0, \quad C_1=0$$

$$\rho_v = 0.3z^2 \cos (10^4 t) \text{ mC/m}^3$$

$$30. \text{ (D) } \mathbf{J}_d = \nabla \times \mathbf{H} = \frac{\partial H_z}{\partial y} \mathbf{u}_x$$

$$= 40 \sin (10^9 t - 4y) \mathbf{u}_y \text{ A/m}$$

$$31. \text{ (A) } \nabla \times \mathbf{H} = -\frac{\partial H_z}{\partial y} \mathbf{u}_z = \epsilon \frac{\partial \mathbf{E}}{\partial t}$$

$$= 6\beta \sin (\beta x) \cos (10^8 t) \mathbf{u}_y$$

$$\mathbf{E} = \frac{1}{\epsilon} \int 6\beta \sin (\beta x) \cos (10^8 t) \mathbf{u}_y dt$$

$$= \frac{6\beta}{\epsilon 10^{10}} \sin (\beta x) \sin (10^8 t) \mathbf{u}_y$$

$$\epsilon = 6.25\epsilon_0, \quad \beta = \frac{\omega}{v} = \frac{\omega}{c} \sqrt{\epsilon_r} = \frac{10^8}{3 \times 10^8} \sqrt{6.25} = 0.833$$

$$\mathbf{E} = \frac{6(0.833)}{6.25\epsilon_0 \times 10^8} \sin (\beta x) \sin (10^8 t) \mathbf{u}_y \text{ V/m}$$

$$= 903 \sin (0.833x) \sin (10^8 t) \mathbf{u}_y \text{ V/m}$$

$$32. \text{ (C) For nonmagnetic medium } \mu_r = 1$$

$$\beta = \frac{\omega}{v} = \frac{\omega}{c} \sqrt{\epsilon_r}, \quad \omega = 10^9, \quad \beta = 8,$$

$$8 = \frac{10^9}{3} \times 10^8 \sqrt{\epsilon_r} \Rightarrow \epsilon_r = 5.76$$
